

Chapter C4

A64 Instruction Set Encoding

This chapter describes the encoding of the A64 instruction set. It contains the following section:

- [A64 instruction set encoding on page C4-210.](#)

In this chapter:

- In the decode tables, an entry of - for a field value means the value of the field does not affect the decoding.
- In the decode diagrams, a shaded field indicates that the bits in that field are not used in that level of decode.

C4.1 A64 instruction set encoding

The A64 instruction encoding is:



Table C4-1 Main encoding table for the A64 instruction set

Decode fields	
op0	Decode group or instruction page
00xx	Unallocated.
100x	<i>Data Processing -- Immediate</i>
101x	<i>Branches, Exception Generating and System instructions on page C4-214</i>
x1x0	<i>Loads and Stores on page C4-219</i>
x101	<i>Data Processing -- Register on page C4-253</i>
x111	<i>Data Processing -- Scalar Floating-Point and Advanced SIMD on page C4-261</i>

C4.1.1 Data Processing -- Immediate

This section describes the encoding of the Data Processing -- Immediate group. This section is decoded from *A64 instruction set encoding*.



Table C4-2 Encoding table for the Data Processing -- Immediate group

Decode fields	
op0	Decode group or instruction page
00x	<i>PC-rel. addressing</i>
01x	<i>Add/subtract (immediate) on page C4-211</i>
100	<i>Logical (immediate) on page C4-211</i>
101	<i>Move wide (immediate) on page C4-212</i>
110	<i>Bitfield on page C4-213</i>
111	<i>Extract on page C4-213</i>

PC-rel. addressing

This section describes the encoding of the PC-rel. addressing instruction class. This section is decoded from *Data Processing -- Immediate*.

31	30	29	28	27	26	25	24	23							5	4		0
opimmlo									immhi							Rd		

Decode fields	Instruction page
op	
0	ADR
1	ADRP

Add/subtract (immediate)

This section describes the encoding of the Add/subtract (immediate) instruction class. This section is decoded from *Data Processing -- Immediate* on page C4-210.

31	30	29	28	27	26	25	24	23	22	21				10	9		5	4		0
sf	op	S	1	0	0	0	1	shift	imm12						Rn			Rd		

Decode fields				Instruction page
sf	op	S	shift	
-	-	-	1x	Unallocated.
0	0	0	-	ADD (immediate) - 32-bit variant on page C6-525
0	0	1	-	ADDS (immediate) - 32-bit variant on page C6-531
0	1	0	-	SUB (immediate) - 32-bit variant on page C6-930
0	1	1	-	SUBS (immediate) - 32-bit variant on page C6-936
1	0	0	-	ADD (immediate) - 64-bit variant on page C6-525
1	0	1	-	ADDS (immediate) - 64-bit variant on page C6-531
1	1	0	-	SUB (immediate) - 64-bit variant on page C6-930
1	1	1	-	SUBS (immediate) - 64-bit variant on page C6-936

Logical (immediate)

This section describes the encoding of the Logical (immediate) instruction class. This section is decoded from *Data Processing -- Immediate* on page C4-210.

31	30	29	28	27	26	25	24	23	22	21			16	15			10	9			5	4			0
sf	opc	1	0	0	1	0	0	N					immr				imms				Rn				Rd

Decode fields			Instruction page
sf	opc	N	
0	-	1	Unallocated.
0	00	0	AND (immediate) - 32-bit variant on page C6-537
0	01	0	ORR (immediate) - 32-bit variant on page C6-790
0	10	0	EOR (immediate) - 32-bit variant on page C6-619
0	11	0	ANDS (immediate) - 32-bit variant on page C6-540
1	00	-	AND (immediate) - 64-bit variant on page C6-537
1	01	-	ORR (immediate) - 64-bit variant on page C6-790
1	10	-	EOR (immediate) - 64-bit variant on page C6-619
1	11	-	ANDS (immediate) - 64-bit variant on page C6-540

Move wide (immediate)

This section describes the encoding of the Move wide (immediate) instruction class. This section is decoded from [Data Processing -- Immediate on page C4-210](#).

31	30	29	28	27	26	25	24	23	22	21	20											5	4			0
sf	opc	1	0	0	1	0	1	hw																		Rd

Decode fields			Instruction page
sf	opc	hw	
-	01	-	Unallocated.
0	-	1x	Unallocated.
0	00	-	MOVN - 32-bit variant on page C6-768
0	10	-	MOVZ - 32-bit variant on page C6-770
0	11	-	MOVK - 32-bit variant on page C6-767
1	00	-	MOVN - 64-bit variant on page C6-768
1	10	-	MOVZ - 64-bit variant on page C6-770
1	11	-	MOVK - 64-bit variant on page C6-767

Bitfield

This section describes the encoding of the Bitfield instruction class. This section is decoded from *Data Processing -- Immediate* on page C4-210.

31	30	29	28	27	26	25	24	23	22	21		16	15		10	9		5	4		0
sf	opc	1	0	0	1	1	0	N				immr			imms			Rn			Rd

Decode fields

Instruction page

sf	opc	N	
-	11	-	Unallocated.
0	-	1	Unallocated.
0	00	0	SBFM - 32-bit variant on page C6-820
0	01	0	BFM - 32-bit variant on page C6-553
0	10	0	UBFM - 32-bit variant on page C6-960
1	-	0	Unallocated.
1	00	1	SBFM - 64-bit variant on page C6-820
1	01	1	BFM - 64-bit variant on page C6-553
1	10	1	UBFM - 64-bit variant on page C6-960

Extract

This section describes the encoding of the Extract instruction class. This section is decoded from *Data Processing -- Immediate* on page C4-210.

31	30	29	28	27	26	25	24	23	22	21	20	16				15	10		9	5		4	0		
sf	op21	1	0	0	1	1	1	N	o0	Rm				imms				Rn				Rd			

Decode fields

Instruction page

sf	op21	N	o0	imms	
-	x1	-	-	-	Unallocated.
-	00	-	1	-	Unallocated.
-	1x	-	-	-	Unallocated.
0	-	-	-	1xxxxx	Unallocated.
0	-	1	-	-	Unallocated.
0	00	0	0	0xxxxx	EXTR - 32-bit variant on page C6-625
1	-	0	-	-	Unallocated.
1	00	1	0	-	EXTR - 64-bit variant on page C6-625

C4.1.2 Branches, Exception Generating and System instructions

This section describes the encoding of the Branches, Exception Generating and System instructions group. This section is decoded from *A64 instruction set encoding* on page C4-210.

31	29 28	25	22 21						0
op0	101	op1							

Table C4-3 Encoding table for the Branches, Exception Generating and System instructions group

Decode fields		Decode group or instruction page
op0	op1	
010	0xxx	<i>Conditional branch (immediate)</i>
010	1xxx	Unallocated.
110	00xx	<i>Exception generation on page C4-215</i>
110	0100	<i>System on page C4-215</i>
110	0101	Unallocated.
110	011x	Unallocated.
110	1xxx	<i>Unconditional branch (register) on page C4-217</i>
x00	-	<i>Unconditional branch (immediate) on page C4-217</i>
x01	0xxx	<i>Compare and branch (immediate) on page C4-218</i>
x01	1xxx	<i>Test and branch (immediate) on page C4-218</i>
x11	-	Unallocated.

Conditional branch (immediate)

This section describes the encoding of the Conditional branch (immediate) instruction class. This section is decoded from *Branches, Exception Generating and System instructions*.

31	30	29	28	27	26	25	24	23					5	4	3	0
0	1	0	1	0	1	0	o1	imm19						o0	cond	

Decode fields		Instruction page
o1	o0	
0	0	<i>B.cond</i>
0	1	Unallocated.
1	-	Unallocated.

Exception generation

This section describes the encoding of the Exception generation instruction class. This section is decoded from *Branches, Exception Generating and System instructions* on page C4-214.

31	30	29	28	27	26	25	24	23	21	20					5	4		2	1	0
1	1	0	1	0	1	0	0	opc	imm16						op2			LL		

Decode fields			Instruction page
opc	op2	LL	
-	xx1	-	Unallocated.
-	x1x	-	Unallocated.
-	1xx	-	Unallocated.
000	000	00	Unallocated.
000	000	01	SVC
000	000	10	HVC
000	000	11	SMC
001	000	x1	Unallocated.
001	000	00	BRK
001	000	1x	Unallocated.
010	000	x1	Unallocated.
010	000	00	HLT
010	000	1x	Unallocated.
011	000	-	Unallocated.
100	000	-	Unallocated.
101	000	00	Unallocated.
101	000	01	DCPS1
101	000	10	DCPS2
101	000	11	DCPS3
11x	000	-	Unallocated.

System

This section describes the encoding of the System instruction class. This section is decoded from *Branches, Exception Generating and System instructions* on page C4-214.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	16	15	12	11	8	7	5	4	0
1	1	0	1	0	1	0	1	0	0	L	op0	op1	CRn	CRm	op2	Rt						

Decode fields							Instruction page	Architecture version
L	op0	op1	CRn	CRm	op2	Rt		
0	00	-	000x	-	-	-	Unallocated.	-
0	00	-	0100	-	-	!= 11111	Unallocated.	-
0	00	-	0100	-	-	11111	MSR (immediate)	-
0	00	-	0101	-	-	-	Unallocated.	-
0	00	-	011x	-	-	-	Unallocated.	-
0	00	-	1xxx	-	-	-	Unallocated.	-
0	00	xx0	001x	-	-	-	Unallocated.	-
0	00	x0x	001x	-	-	-	Unallocated.	-
0	00	011	001x	-	-	!= 11111	Unallocated.	-
0	00	011	0010	!= 00x0	-	11111	HINT - Hints 8 to 15, and 24 to 127 variant on page C6-627	-
0	00	011	0010	0000	000	11111	NOP	-
0	00	011	0010	0000	001	11111	YIELD	-
0	00	011	0010	0000	010	11111	WFE	-
0	00	011	0010	0000	011	11111	WFI	-
0	00	011	0010	0000	100	11111	SEV	-
0	00	011	0010	0000	101	11111	SEVL	-
0	00	011	0010	0000	11x	11111	HINT - Hints 6 and 7 variant on page C6-627	-
0	00	011	0010	0010	!= 00x	11111	HINT - Hints 17 to 23 variant on page C6-627	-
0	00	011	0010	0010	000	11111	ESB	ARMv8.2
0	00	011	0010	0010	001	11111	PSB CSYNC	ARMv8.2
0	00	011	0011	-	000	-	Unallocated.	-
0	00	011	0011	-	001	-	Unallocated.	-
0	00	011	0011	-	010	11111	CLREX	-
0	00	011	0011	-	011	-	Unallocated.	-
0	00	011	0011	-	100	11111	DSB	-
0	00	011	0011	-	101	11111	DMB	-
0	00	011	0011	-	110	11111	ISB	-

Decode fields							Instruction page	Architecture version
L	op0	op1	CRn	CRm	op2	Rt		
0	00	011	0011	-	111	-	Unallocated.	-
0	00	1xx	001x	-	-	-	Unallocated.	-
0	01	-	-	-	-	-	SYS	-
0	1x	-	-	-	-	-	MSR (register)	-
1	00	-	-	-	-	-	Unallocated.	-
1	01	-	-	-	-	-	SYSL	-
1	1x	-	-	-	-	-	MRS	-

Unconditional branch (register)

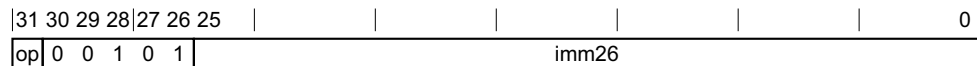
This section describes the encoding of the Unconditional branch (register) instruction class. This section is decoded from *Branches, Exception Generating and System instructions on page C4-214*.

31	30	29	28	27	26	25	24	21	20	16	15	10	9	5	4	0
1	1	0	1	0	1	1		opc		op2		op3		Rn		op4

Decode fields					Instruction page
opc	op2	op3	Rn	op4	
-	-	-	-	!= 00000	Unallocated.
-	-	!= 000000	-	-	Unallocated.
-	!= 11111	-	-	-	Unallocated.
0000	11111	000000	-	00000	BR
0001	11111	000000	-	00000	BLR
0010	11111	000000	-	00000	RET
0011	-	-	-	-	Unallocated.
010x	-	-	!= 11111	-	Unallocated.
0100	11111	000000	11111	00000	ERET
0101	11111	000000	11111	00000	DRPS
011x	-	-	-	-	Unallocated.
1xxx	-	-	-	-	Unallocated.

Unconditional branch (immediate)

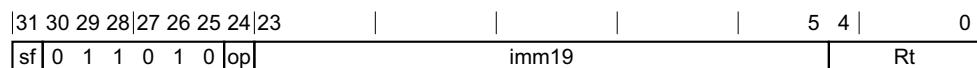
This section describes the encoding of the Unconditional branch (immediate) instruction class. This section is decoded from *Branches, Exception Generating and System instructions on page C4-214*.



Decode fields	Instruction page
op	
0	B
1	BL

Compare and branch (immediate)

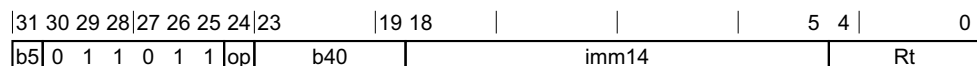
This section describes the encoding of the Compare and branch (immediate) instruction class. This section is decoded from *Branches, Exception Generating and System instructions* on page C4-214.



Decode fields		Instruction page
sf	op	
0	0	CBZ - 32-bit variant on page C6-574
0	1	CBNZ - 32-bit variant on page C6-573
1	0	CBZ - 64-bit variant on page C6-574
1	1	CBNZ - 64-bit variant on page C6-573

Test and branch (immediate)

This section describes the encoding of the Test and branch (immediate) instruction class. This section is decoded from *Branches, Exception Generating and System instructions* on page C4-214.



Decode fields	Instruction page
op	
0	TBZ
1	TBNZ

C4.1.3 Loads and Stores

This section describes the encoding of the Loads and Stores group. This section is decoded from *A64 instruction set encoding* on page C4-210.



Table C4-4 Encoding table for the Loads and Stores group

Decode fields						Decode group or instruction page
op0	op1	op2	op3	op4	op5	
0	00	1	00	000000	-	Advanced SIMD load/store multiple structures on page C4-220
0	00	1	01	0xxxxx	-	Advanced SIMD load/store multiple structures (post-indexed) on page C4-221
0	00	1	0x	1xxxxx	-	Unallocated.
0	00	1	10	x00000	-	Advanced SIMD load/store single structure on page C4-222
0	00	1	11	-	-	Advanced SIMD load/store single structure (post-indexed) on page C4-225
0	00	1	x0	x1xxxx	-	Unallocated.
0	00	1	x0	xx1xxx	-	Unallocated.
0	00	1	x0	xxx1xx	-	Unallocated.
0	00	1	x0	xxxx1x	-	Unallocated.
0	00	1	x0	xxxxx1	-	Unallocated.
1	00	1	-	-	-	Unallocated.
-	00	0	0x	-	-	Load/store exclusive on page C4-229
-	00	0	1x	-	-	Unallocated.
-	01	-	0x	-	-	Load register (literal) on page C4-231
-	01	-	1x	-	-	Unallocated.
-	10	-	00	-	-	Load/store no-allocate pair (offset) on page C4-232
-	10	-	01	-	-	Load/store register pair (post-indexed) on page C4-233
-	10	-	10	-	-	Load/store register pair (offset) on page C4-233
-	10	-	11	-	-	Load/store register pair (pre-indexed) on page C4-234
-	11	-	0x	0xxxxx	00	Load/store register (unscaled immediate) on page C4-235
-	11	-	0x	0xxxxx	01	Load/store register (immediate post-indexed) on page C4-236
-	11	-	0x	0xxxxx	10	Load/store register (unprivileged) on page C4-237
-	11	-	0x	0xxxxx	11	Load/store register (immediate pre-indexed) on page C4-238

Table C4-4 Encoding table for the Loads and Stores group (continued)

Decode fields						Decode group or instruction page
op0	op1	op2	op3	op4	op5	
-	11	-	0x	1xxxxx	00	Atomic memory operations on page C4-239
-	11	-	0x	1xxxxx	10	Load/store register (register offset) on page C4-250
-	11	-	1x	-	-	Load/store register (unsigned immediate) on page C4-251

Advanced SIMD load/store multiple structures

This section describes the encoding of the Advanced SIMD load/store multiple structures instruction class. This section is decoded from [Loads and Stores on page C4-219](#).

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	12	11	10	9	5	4	0
0	Q	0	0	1	1	0	0	0	L	0	0	0	0	0	0	opcode	size	Rn					Rt

Decode fields		Instruction page
L	opcode	
0	0000	ST4 (multiple structures)
0	0001	Unallocated.
0	0010	ST1 (multiple structures) - Four registers variant on page C7-1597
0	0011	Unallocated.
0	0100	ST3 (multiple structures)
0	0101	Unallocated.
0	0110	ST1 (multiple structures) - Three registers variant on page C7-1597
0	0111	ST1 (multiple structures) - One register variant on page C7-1597
0	1000	ST2 (multiple structures)
0	1001	Unallocated.
0	1010	ST1 (multiple structures) - Two registers variant on page C7-1597
0	1011	Unallocated.
0	11xx	Unallocated.
1	0000	LD4 (multiple structures)
1	0001	Unallocated.
1	0010	LD1 (multiple structures) - Four registers variant on page C7-1312
1	0011	Unallocated.
1	0100	LD3 (multiple structures)
1	0101	Unallocated.

Decode fields		Instruction page	
L	opcode		
1	0110		LD1 (multiple structures) - <i>Three registers variant</i> on page C7-1312
1	0111		LD1 (multiple structures) - <i>One register variant</i> on page C7-1312
1	1000		LD2 (multiple structures)
1	1001		Unallocated.
1	1010		LD1 (multiple structures) - <i>Two registers variant</i> on page C7-1312
1	1011		Unallocated.
1	11xx		Unallocated.

Advanced SIMD load/store multiple structures (post-indexed)

This section describes the encoding of the Advanced SIMD load/store multiple structures (post-indexed) instruction class. This section is decoded from *Loads and Stores* on page C4-219.

31	30	29	28	27	26	25	24	23	22	21	20	16	15	12	11	10	9	5	4	0
0	Q	0	0	1	1	0	0	1	L	0		Rm		opcode	size		Rn			Rt

Decode fields			Instruction page
L	Rm	opcode	
0	-	0001	Unallocated.
0	-	0011	Unallocated.
0	-	0101	Unallocated.
0	-	1001	Unallocated.
0	-	1011	Unallocated.
0	-	11xx	Unallocated.
0	!= 11111	0000	ST4 (multiple structures) - <i>Register offset variant</i> on page C7-1619
0	!= 11111	0010	ST1 (multiple structures) - <i>Four registers, register offset variant</i> on page C7-1598
0	!= 11111	0100	ST3 (multiple structures) - <i>Register offset variant</i> on page C7-1612
0	!= 11111	0110	ST1 (multiple structures) - <i>Three registers, register offset variant</i> on page C7-1598
0	!= 11111	0111	ST1 (multiple structures) - <i>One register, register offset variant</i> on page C7-1597
0	!= 11111	1000	ST2 (multiple structures) - <i>Register offset variant</i> on page C7-1605
0	!= 11111	1010	ST1 (multiple structures) - <i>Two registers, register offset variant</i> on page C7-1598
0	11111	0000	ST4 (multiple structures) - <i>Immediate offset variant</i> on page C7-1619
0	11111	0010	ST1 (multiple structures) - <i>Four registers, immediate offset variant</i> on page C7-1598
0	11111	0100	ST3 (multiple structures) - <i>Immediate offset variant</i> on page C7-1612

Decode fields			Instruction page
L	Rm	opcode	
0	11111	0110	ST1 (multiple structures) - <i>Three registers, immediate offset variant on page C7-1598</i>
0	11111	0111	ST1 (multiple structures) - <i>One register, immediate offset variant on page C7-1597</i>
0	11111	1000	ST2 (multiple structures) - <i>Immediate offset variant on page C7-1605</i>
0	11111	1010	ST1 (multiple structures) - <i>Two registers, immediate offset variant on page C7-1598</i>
1	-	0001	Unallocated.
1	-	0011	Unallocated.
1	-	0101	Unallocated.
1	-	1001	Unallocated.
1	-	1011	Unallocated.
1	-	11xx	Unallocated.
1	!= 11111	0000	LD4 (multiple structures) - <i>Register offset variant on page C7-1343</i>
1	!= 11111	0010	LD1 (multiple structures) - <i>Four registers, register offset variant on page C7-1313</i>
1	!= 11111	0100	LD3 (multiple structures) - <i>Register offset variant on page C7-1333</i>
1	!= 11111	0110	LD1 (multiple structures) - <i>Three registers, register offset variant on page C7-1313</i>
1	!= 11111	0111	LD1 (multiple structures) - <i>One register, register offset variant on page C7-1312</i>
1	!= 11111	1000	LD2 (multiple structures) - <i>Register offset variant on page C7-1323</i>
1	!= 11111	1010	LD1 (multiple structures) - <i>Two registers, register offset variant on page C7-1313</i>
1	11111	0000	LD4 (multiple structures) - <i>Immediate offset variant on page C7-1343</i>
1	11111	0010	LD1 (multiple structures) - <i>Four registers, immediate offset variant on page C7-1313</i>
1	11111	0100	LD3 (multiple structures) - <i>Immediate offset variant on page C7-1333</i>
1	11111	0110	LD1 (multiple structures) - <i>Three registers, immediate offset variant on page C7-1313</i>
1	11111	0111	LD1 (multiple structures) - <i>One register, immediate offset variant on page C7-1312</i>
1	11111	1000	LD2 (multiple structures) - <i>Immediate offset variant on page C7-1323</i>
1	11111	1010	LD1 (multiple structures) - <i>Two registers, immediate offset variant on page C7-1313</i>

Advanced SIMD load/store single structure

This section describes the encoding of the Advanced SIMD load/store single structure instruction class. This section is decoded from *Loads and Stores* on page C4-219.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	13	12	11	10	9	5	4	0
0	Q	0	0	1	1	0	1	0	L	R	0	0	0	0	0	opcode	S	size	Rn					Rt

Decode fields						Instruction page
L	R	opcode	S	size		
0	-	11x	-	-		Unallocated.
0	0	000	-	-		ST1 (single structure) - 8-bit variant on page C7-1601
0	0	001	-	-		ST3 (single structure) - 8-bit variant on page C7-1615
0	0	010	-	x0		ST1 (single structure) - 16-bit variant on page C7-1601
0	0	010	-	x1		Unallocated.
0	0	011	-	x0		ST3 (single structure) - 16-bit variant on page C7-1615
0	0	011	-	x1		Unallocated.
0	0	100	-	00		ST1 (single structure) - 32-bit variant on page C7-1601
0	0	100	-	1x		Unallocated.
0	0	100	0	01		ST1 (single structure) - 64-bit variant on page C7-1601
0	0	100	1	01		Unallocated.
0	0	101	-	00		ST3 (single structure) - 32-bit variant on page C7-1615
0	0	101	-	10		Unallocated.
0	0	101	0	01		ST3 (single structure) - 64-bit variant on page C7-1615
0	0	101	0	11		Unallocated.
0	0	101	1	x1		Unallocated.
0	1	000	-	-		ST2 (single structure) - 8-bit variant on page C7-1608
0	1	001	-	-		ST4 (single structure) - 8-bit variant on page C7-1622
0	1	010	-	x0		ST2 (single structure) - 16-bit variant on page C7-1608
0	1	010	-	x1		Unallocated.
0	1	011	-	x0		ST4 (single structure) - 16-bit variant on page C7-1622
0	1	011	-	x1		Unallocated.
0	1	100	-	00		ST2 (single structure) - 32-bit variant on page C7-1608
0	1	100	-	10		Unallocated.
0	1	100	0	01		ST2 (single structure) - 64-bit variant on page C7-1608
0	1	100	0	11		Unallocated.
0	1	100	1	x1		Unallocated.
0	1	101	-	00		ST4 (single structure) - 32-bit variant on page C7-1622

Decode fields					Instruction page
L	R	opcode	S	size	
0	1	101	-	10	Unallocated.
0	1	101	0	01	ST4 (single structure) - 64-bit variant on page C7-1622
0	1	101	0	11	Unallocated.
0	1	101	1	x1	Unallocated.
1	0	000	-	-	LD1 (single structure) - 8-bit variant on page C7-1316
1	0	001	-	-	LD3 (single structure) - 8-bit variant on page C7-1336
1	0	010	-	x0	LD1 (single structure) - 16-bit variant on page C7-1316
1	0	010	-	x1	Unallocated.
1	0	011	-	x0	LD3 (single structure) - 16-bit variant on page C7-1336
1	0	011	-	x1	Unallocated.
1	0	100	-	00	LD1 (single structure) - 32-bit variant on page C7-1316
1	0	100	-	1x	Unallocated.
1	0	100	0	01	LD1 (single structure) - 64-bit variant on page C7-1316
1	0	100	1	01	Unallocated.
1	0	101	-	00	LD3 (single structure) - 32-bit variant on page C7-1336
1	0	101	-	10	Unallocated.
1	0	101	0	01	LD3 (single structure) - 64-bit variant on page C7-1336
1	0	101	0	11	Unallocated.
1	0	101	1	x1	Unallocated.
1	0	110	0	-	LD1R
1	0	110	1	-	Unallocated.
1	0	111	0	-	LD3R
1	0	111	1	-	Unallocated.
1	1	000	-	-	LD2 (single structure) - 8-bit variant on page C7-1326
1	1	001	-	-	LD4 (single structure) - 8-bit variant on page C7-1346
1	1	010	-	x0	LD2 (single structure) - 16-bit variant on page C7-1326
1	1	010	-	x1	Unallocated.
1	1	011	-	x0	LD4 (single structure) - 16-bit variant on page C7-1346
1	1	011	-	x1	Unallocated.
1	1	100	-	00	LD2 (single structure) - 32-bit variant on page C7-1326
1	1	100	-	10	Unallocated.
1	1	100	0	01	LD2 (single structure) - 64-bit variant on page C7-1326

Decode fields					Instruction page	
L	R	opcode	S	size		
1	1	100	0	11		Unallocated.
1	1	100	1	x1		Unallocated.
1	1	101	-	00		LD4 (single structure) - 32-bit variant on page C7-1346
1	1	101	-	10		Unallocated.
1	1	101	0	01		LD4 (single structure) - 64-bit variant on page C7-1346
1	1	101	0	11		Unallocated.
1	1	101	1	x1		Unallocated.
1	1	110	0	-		LD2R
1	1	110	1	-		Unallocated.
1	1	111	0	-		LD4R
1	1	111	1	-		Unallocated.

Advanced SIMD load/store single structure (post-indexed)

This section describes the encoding of the Advanced SIMD load/store single structure (post-indexed) instruction class. This section is decoded from [Loads and Stores on page C4-219](#).

31	30	29	28	27	26	25	24	23	22	21	20	16	15	13	12	11	10	9	5	4	0
0	Q	0	0	1	1	0	1	1	L	R		Rm	opcode	S	size	Rn					Rt

Decode fields						Instruction page
L	R	Rm	opcode	S	size	
0	-	-	11x	-	-	Unallocated.
0	0	-	010	-	x1	Unallocated.
0	0	-	011	-	x1	Unallocated.
0	0	-	100	-	1x	Unallocated.
0	0	-	100	1	01	Unallocated.
0	0	-	101	-	10	Unallocated.
0	0	-	101	0	11	Unallocated.
0	0	-	101	1	x1	Unallocated.
0	0	!= 11111	000	-	-	ST1 (single structure) - 8-bit, register offset variant on page C7-1601
0	0	!= 11111	001	-	-	ST3 (single structure) - 8-bit, register offset variant on page C7-1615
0	0	!= 11111	010	-	x0	ST1 (single structure) - 16-bit, register offset variant on page C7-1602
0	0	!= 11111	011	-	x0	ST3 (single structure) - 16-bit, register offset variant on page C7-1616

Decode fields						Instruction page
L	R	Rm	opcode	S	size	
0	0	!= 11111	100	-	00	ST1 (single structure) - 32-bit, register offset variant on page C7-1602
0	0	!= 11111	100	0	01	ST1 (single structure) - 64-bit, register offset variant on page C7-1602
0	0	!= 11111	101	-	00	ST3 (single structure) - 32-bit, register offset variant on page C7-1616
0	0	!= 11111	101	0	01	ST3 (single structure) - 64-bit, register offset variant on page C7-1616
0	0	11111	000	-	-	ST1 (single structure) - 8-bit, immediate offset variant on page C7-1601
0	0	11111	001	-	-	ST3 (single structure) - 8-bit, immediate offset variant on page C7-1615
0	0	11111	010	-	x0	ST1 (single structure) - 16-bit, immediate offset variant on page C7-1602
0	0	11111	011	-	x0	ST3 (single structure) - 16-bit, immediate offset variant on page C7-1616
0	0	11111	100	-	00	ST1 (single structure) - 32-bit, immediate offset variant on page C7-1602
0	0	11111	100	0	01	ST1 (single structure) - 64-bit, immediate offset variant on page C7-1602
0	0	11111	101	-	00	ST3 (single structure) - 32-bit, immediate offset variant on page C7-1616
0	0	11111	101	0	01	ST3 (single structure) - 64-bit, immediate offset variant on page C7-1616
0	1	-	010	-	x1	Unallocated.
0	1	-	011	-	x1	Unallocated.
0	1	-	100	-	10	Unallocated.
0	1	-	100	0	11	Unallocated.
0	1	-	100	1	x1	Unallocated.
0	1	-	101	-	10	Unallocated.
0	1	-	101	0	11	Unallocated.
0	1	-	101	1	x1	Unallocated.
0	1	!= 11111	000	-	-	ST2 (single structure) - 8-bit, register offset variant on page C7-1608
0	1	!= 11111	001	-	-	ST4 (single structure) - 8-bit, register offset variant on page C7-1622
0	1	!= 11111	010	-	x0	ST2 (single structure) - 16-bit, register offset variant on page C7-1609
0	1	!= 11111	011	-	x0	ST4 (single structure) - 16-bit, register offset variant on page C7-1623
0	1	!= 11111	100	-	00	ST2 (single structure) - 32-bit, register offset variant on page C7-1609
0	1	!= 11111	100	0	01	ST2 (single structure) - 64-bit, register offset variant on page C7-1609
0	1	!= 11111	101	-	00	ST4 (single structure) - 32-bit, register offset variant on page C7-1623
0	1	!= 11111	101	0	01	ST4 (single structure) - 64-bit, register offset variant on page C7-1623
0	1	11111	000	-	-	ST2 (single structure) - 8-bit, immediate offset variant on page C7-1608
0	1	11111	001	-	-	ST4 (single structure) - 8-bit, immediate offset variant on page C7-1622
0	1	11111	010	-	x0	ST2 (single structure) - 16-bit, immediate offset variant on page C7-1609
0	1	11111	011	-	x0	ST4 (single structure) - 16-bit, immediate offset variant on page C7-1623

Decode fields						Instruction page
L	R	Rm	opcode	S	size	
0	1	11111	100	-	00	ST2 (single structure) - 32-bit, immediate offset variant on page C7-1609
0	1	11111	100	0	01	ST2 (single structure) - 64-bit, immediate offset variant on page C7-1609
0	1	11111	101	-	00	ST4 (single structure) - 32-bit, immediate offset variant on page C7-1623
0	1	11111	101	0	01	ST4 (single structure) - 64-bit, immediate offset variant on page C7-1623
1	0	-	010	-	x1	Unallocated.
1	0	-	011	-	x1	Unallocated.
1	0	-	100	-	1x	Unallocated.
1	0	-	100	1	01	Unallocated.
1	0	-	101	-	10	Unallocated.
1	0	-	101	0	11	Unallocated.
1	0	-	101	1	x1	Unallocated.
1	0	-	110	1	-	Unallocated.
1	0	-	111	1	-	Unallocated.
1	0	!= 11111	000	-	-	LD1 (single structure) - 8-bit, register offset variant on page C7-1316
1	0	!= 11111	001	-	-	LD3 (single structure) - 8-bit, register offset variant on page C7-1336
1	0	!= 11111	010	-	x0	LD1 (single structure) - 16-bit, register offset variant on page C7-1317
1	0	!= 11111	011	-	x0	LD3 (single structure) - 16-bit, register offset variant on page C7-1337
1	0	!= 11111	100	-	00	LD1 (single structure) - 32-bit, register offset variant on page C7-1317
1	0	!= 11111	100	0	01	LD1 (single structure) - 64-bit, register offset variant on page C7-1317
1	0	!= 11111	101	-	00	LD3 (single structure) - 32-bit, register offset variant on page C7-1337
1	0	!= 11111	101	0	01	LD3 (single structure) - 64-bit, register offset variant on page C7-1337
1	0	!= 11111	110	0	-	LD1R - Register offset variant on page C7-1320
1	0	!= 11111	111	0	-	LD3R - Register offset variant on page C7-1340
1	0	11111	000	-	-	LD1 (single structure) - 8-bit, immediate offset variant on page C7-1316
1	0	11111	001	-	-	LD3 (single structure) - 8-bit, immediate offset variant on page C7-1336
1	0	11111	010	-	x0	LD1 (single structure) - 16-bit, immediate offset variant on page C7-1317
1	0	11111	011	-	x0	LD3 (single structure) - 16-bit, immediate offset variant on page C7-1337
1	0	11111	100	-	00	LD1 (single structure) - 32-bit, immediate offset variant on page C7-1317
1	0	11111	100	0	01	LD1 (single structure) - 64-bit, immediate offset variant on page C7-1317
1	0	11111	101	-	00	LD3 (single structure) - 32-bit, immediate offset variant on page C7-1337
1	0	11111	101	0	01	LD3 (single structure) - 64-bit, immediate offset variant on page C7-1337
1	0	11111	110	0	-	LD1R - Immediate offset variant on page C7-1320

Decode fields			Instruction page			
L	R	Rm	opcode	S	size	
1	0	11111	111	0	-	LD3R - Immediate offset variant on page C7-1340
1	1	-	010	-	x1	Unallocated.
1	1	-	011	-	x1	Unallocated.
1	1	-	100	-	10	Unallocated.
1	1	-	100	0	11	Unallocated.
1	1	-	100	1	x1	Unallocated.
1	1	-	101	-	10	Unallocated.
1	1	-	101	0	11	Unallocated.
1	1	-	101	1	x1	Unallocated.
1	1	-	110	1	-	Unallocated.
1	1	-	111	1	-	Unallocated.
1	1	!= 11111	000	-	-	LD2 (single structure) - 8-bit, register offset variant on page C7-1326
1	1	!= 11111	001	-	-	LD4 (single structure) - 8-bit, register offset variant on page C7-1346
1	1	!= 11111	010	-	x0	LD2 (single structure) - 16-bit, register offset variant on page C7-1327
1	1	!= 11111	011	-	x0	LD4 (single structure) - 16-bit, register offset variant on page C7-1347
1	1	!= 11111	100	-	00	LD2 (single structure) - 32-bit, register offset variant on page C7-1327
1	1	!= 11111	100	0	01	LD2 (single structure) - 64-bit, register offset variant on page C7-1327
1	1	!= 11111	101	-	00	LD4 (single structure) - 32-bit, register offset variant on page C7-1347
1	1	!= 11111	101	0	01	LD4 (single structure) - 64-bit, register offset variant on page C7-1347
1	1	!= 11111	110	0	-	LD2R - Register offset variant on page C7-1330
1	1	!= 11111	111	0	-	LD4R - Register offset variant on page C7-1350
1	1	11111	000	-	-	LD2 (single structure) - 8-bit, immediate offset variant on page C7-1326
1	1	11111	001	-	-	LD4 (single structure) - 8-bit, immediate offset variant on page C7-1346
1	1	11111	010	-	x0	LD2 (single structure) - 16-bit, immediate offset variant on page C7-1327
1	1	11111	011	-	x0	LD4 (single structure) - 16-bit, immediate offset variant on page C7-1347
1	1	11111	100	-	00	LD2 (single structure) - 32-bit, immediate offset variant on page C7-1327
1	1	11111	100	0	01	LD2 (single structure) - 64-bit, immediate offset variant on page C7-1327
1	1	11111	101	-	00	LD4 (single structure) - 32-bit, immediate offset variant on page C7-1347
1	1	11111	101	0	01	LD4 (single structure) - 64-bit, immediate offset variant on page C7-1347
1	1	11111	110	0	-	LD2R - Immediate offset variant on page C7-1330
1	1	11111	111	0	-	LD4R - Immediate offset variant on page C7-1350

Load/store exclusive

This section describes the encoding of the Load/store exclusive instruction class. This section is decoded from *Loads and Stores* on page C4-219.

31	30	29	28	27	26	25	24	23	22	21	20	16	15	14	10	9	5	4	0
size	0	0	1	0	0	0	0	o2	L	o1		Rs	o0	Rt2		Rn		Rt	

Decode fields						Instruction page	Architecture version
size	o2	L	o1	o0	Rt2		
-	1	-	1	-	!= 11111	Unallocated.	-
0x	0	-	1	-	!= 11111	Unallocated.	-
00	0	0	0	0	-	STXRB	-
00	0	0	0	1	-	STLXRB	-
00	0	0	1	0	11111	CASP, CASPA, CASPAL, CASPL - 32-bit, no memory ordering variant on page C6-568	ARMv8.1
00	0	0	1	1	11111	CASP, CASPA, CASPAL, CASPL - 32-bit, release variant on page C6-568	ARMv8.1
00	0	1	0	0	-	LDXRB	-
00	0	1	0	1	-	LDAXRB	-
00	0	1	1	0	11111	CASP, CASPA, CASPAL, CASPL - 32-bit, acquire variant on page C6-568	ARMv8.1
00	0	1	1	1	11111	CASP, CASPA, CASPAL, CASPL - 32-bit, acquire and release variant on page C6-568	ARMv8.1
00	1	0	0	0	-	STLLRB	ARMv8.1
00	1	0	0	1	-	STLRB	-
00	1	0	1	0	11111	CASB, CASAB, CASALB, CASLB - No memory ordering variant on page C6-564	ARMv8.1
00	1	0	1	1	11111	CASB, CASAB, CASALB, CASLB - Release variant on page C6-564	ARMv8.1
00	1	1	0	0	-	LDLARB	ARMv8.1
00	1	1	0	1	-	LDARB	-
00	1	1	1	0	11111	CASB, CASAB, CASALB, CASLB - Acquire variant on page C6-564	ARMv8.1
00	1	1	1	1	11111	CASB, CASAB, CASALB, CASLB - Acquire and release variant on page C6-564	ARMv8.1
01	0	0	0	0	-	STXRH	-
01	0	0	0	1	-	STLXRH	-
01	0	0	1	0	11111	CASP, CASPA, CASPAL, CASPL - 64-bit, no memory ordering variant on page C6-569	ARMv8.1

Decode fields						Instruction page	Architecture version
size	o2	L	o1	o0	Rt2		
01	0	0	1	1	11111	CASP, CASPA, CASPAL, CASPL - <i>64-bit, release variant on page C6-569</i>	ARMv8.1
01	0	1	0	0	-	LDXRH	-
01	0	1	0	1	-	LDAXRH	-
01	0	1	1	0	11111	CASP, CASPA, CASPAL, CASPL - <i>64-bit, acquire variant on page C6-568</i>	ARMv8.1
01	0	1	1	1	11111	CASP, CASPA, CASPAL, CASPL - <i>64-bit, acquire and release variant on page C6-568</i>	ARMv8.1
01	1	0	0	0	-	STLLRH	ARMv8.1
01	1	0	0	1	-	STLRH	-
01	1	0	1	0	11111	CASH, CASAH, CASALH, CASLH - <i>No memory ordering variant on page C6-566</i>	ARMv8.1
01	1	0	1	1	11111	CASH, CASAH, CASALH, CASLH - <i>Release variant on page C6-566</i>	ARMv8.1
01	1	1	0	0	-	LDLARH	ARMv8.1
01	1	1	0	1	-	LDARH	-
01	1	1	1	0	11111	CASH, CASAH, CASALH, CASLH - <i>Acquire variant on page C6-566</i>	ARMv8.1
01	1	1	1	1	11111	CASH, CASAH, CASALH, CASLH - <i>Acquire and release variant on page C6-566</i>	ARMv8.1
10	0	0	0	0	-	STXR - <i>32-bit variant on page C6-922</i>	-
10	0	0	0	1	-	STLXR - <i>32-bit variant on page C6-859</i>	-
10	0	0	1	0	-	STXP - <i>32-bit variant on page C6-920</i>	-
10	0	0	1	1	-	STLXP - <i>32-bit variant on page C6-856</i>	-
10	0	1	0	0	-	LDXR - <i>32-bit variant on page C6-750</i>	-
10	0	1	0	1	-	LDAXR - <i>32-bit variant on page C6-643</i>	-
10	0	1	1	0	-	LDXP - <i>32-bit variant on page C6-748</i>	-
10	0	1	1	1	-	LDAXP - <i>32-bit variant on page C6-641</i>	-
10	1	0	0	0	-	STLLR - <i>32-bit variant on page C6-852</i>	ARMv8.1
10	1	0	0	1	-	STLR - <i>32-bit variant on page C6-853</i>	-
10	1	0	1	0	11111	CAS, CASA, CASAL, CASL - <i>32-bit, no memory ordering variant on page C6-571</i>	ARMv8.1
10	1	0	1	1	11111	CAS, CASA, CASAL, CASL - <i>32-bit, release variant on page C6-571</i>	ARMv8.1
10	1	1	0	0	-	LDLAR - <i>32-bit variant on page C6-661</i>	ARMv8.1

Decode fields						Instruction page	Architecture version
size	o2	L	o1	o0	Rt2		
10	1	1	0	1	-	LDAR - 32-bit variant on page C6-638	-
10	1	1	1	0	11111	CAS, CASA, CASAL, CASL - 32-bit, acquire variant on page C6-571	ARMv8.1
10	1	1	1	1	11111	CAS, CASA, CASAL, CASL - 32-bit, acquire and release variant on page C6-571	ARMv8.1
11	0	0	0	0	-	STXR - 64-bit variant on page C6-922	-
11	0	0	0	1	-	STLXR - 64-bit variant on page C6-859	-
11	0	0	1	0	-	STXP - 64-bit variant on page C6-920	-
11	0	0	1	1	-	STLXP - 64-bit variant on page C6-856	-
11	0	1	0	0	-	LDXR - 64-bit variant on page C6-750	-
11	0	1	0	1	-	LDAXR - 64-bit variant on page C6-643	-
11	0	1	1	0	-	LDXP - 64-bit variant on page C6-748	-
11	0	1	1	1	-	LDAXP - 64-bit variant on page C6-641	-
11	1	0	0	0	-	STLLR - 64-bit variant on page C6-852	ARMv8.1
11	1	0	0	1	-	STLR - 64-bit variant on page C6-853	-
11	1	0	1	0	11111	CAS, CASA, CASAL, CASL - 64-bit, no memory ordering variant on page C6-572	ARMv8.1
11	1	0	1	1	11111	CAS, CASA, CASAL, CASL - 64-bit, release variant on page C6-572	ARMv8.1
11	1	1	0	0	-	LDLAR - 64-bit variant on page C6-661	ARMv8.1
11	1	1	0	1	-	LDAR - 64-bit variant on page C6-638	-
11	1	1	1	0	11111	CAS, CASA, CASAL, CASL - 64-bit, acquire variant on page C6-571	ARMv8.1
11	1	1	1	1	11111	CAS, CASA, CASAL, CASL - 64-bit, acquire and release variant on page C6-571	ARMv8.1

Load register (literal)

This section describes the encoding of the Load register (literal) instruction class. This section is decoded from *Loads and Stores* on page C4-219.

31	30	29	28	27	26	25	24	23								5	4		0
opc	0	1	1	V	0	0			imm19										Rt

Decode fields

Instruction page

opc	V	
00	0	LDR (literal) - 32-bit variant on page C6-673
00	1	LDR (literal, SIMD&FP) - 32-bit variant on page C7-1362
01	0	LDR (literal) - 64-bit variant on page C6-673
01	1	LDR (literal, SIMD&FP) - 64-bit variant on page C7-1362
10	0	LDRSW (literal)
10	1	LDR (literal, SIMD&FP) - 128-bit variant on page C7-1362
11	0	PRFM (literal)
11	1	Unallocated.

Load/store no-allocate pair (offset)

This section describes the encoding of the Load/store no-allocate pair (offset) instruction class. This section is decoded from [Loads and Stores on page C4-219](#).

31	30	29	28	27	26	25	24	23	22	21			15	14		10	9		5	4		0
opc	1	0	1	V	0	0	0	L	imm7				Rt2		Rn		Rt					

Decode fields

Instruction page

opc	V	L	
00	0	0	STNP - 32-bit variant on page C6-865
00	0	1	LDNP - 32-bit variant on page C6-662
00	1	0	STNP (SIMD&FP) - 32-bit variant on page C7-1626
00	1	1	LDNP (SIMD&FP) - 32-bit variant on page C7-1353
01	0	-	Unallocated.
01	1	0	STNP (SIMD&FP) - 64-bit variant on page C7-1626
01	1	1	LDNP (SIMD&FP) - 64-bit variant on page C7-1353
10	0	0	STNP - 64-bit variant on page C6-865
10	0	1	LDNP - 64-bit variant on page C6-662

Decode fields			Instruction page
opc	V	L	
10	1	0	STNP (SIMD&FP) - 128-bit variant on page C7-1626
10	1	1	LDNP (SIMD&FP) - 128-bit variant on page C7-1353
11	-	-	Unallocated.

Load/store register pair (post-indexed)

This section describes the encoding of the Load/store register pair (post-indexed) instruction class. This section is decoded from [Loads and Stores on page C4-219](#).

31	30	29	28	27	26	25	24	23	22	21			15	14		10	9			5	4			0
opc	1	0	1	V	0	0	1	L	imm7				Rt2		Rn			Rt						

Decode fields			Instruction page
opc	V	L	
00	0	0	STP - 32-bit variant on page C6-867
00	0	1	LDP - 32-bit variant on page C6-664
00	1	0	STP (SIMD&FP) - 32-bit variant on page C7-1628
00	1	1	LDP (SIMD&FP) - 32-bit variant on page C7-1355
01	0	0	Unallocated.
01	0	1	LDPSW
01	1	0	STP (SIMD&FP) - 64-bit variant on page C7-1628
01	1	1	LDP (SIMD&FP) - 64-bit variant on page C7-1355
10	0	0	STP - 64-bit variant on page C6-867
10	0	1	LDP - 64-bit variant on page C6-664
10	1	0	STP (SIMD&FP) - 128-bit variant on page C7-1628
10	1	1	LDP (SIMD&FP) - 128-bit variant on page C7-1355
11	-	-	Unallocated.

Load/store register pair (offset)

This section describes the encoding of the Load/store register pair (offset) instruction class. This section is decoded from [Loads and Stores on page C4-219](#).

31	30	29	28	27	26	25	24	23	22	21				15	14			10	9			5	4			0
opc	1	0	1	V	0	1	0	L						imm7				Rt2				Rn				Rt

Decode fields

Instruction page

opc	V	L	
00	0	0	STP - 32-bit variant on page C6-868
00	0	1	LDP - 32-bit variant on page C6-665
00	1	0	STP (SIMD&FP) - 32-bit variant on page C7-1629
00	1	1	LDP (SIMD&FP) - 32-bit variant on page C7-1356
01	0	0	Unallocated.
01	0	1	LDPSW
01	1	0	STP (SIMD&FP) - 64-bit variant on page C7-1629
01	1	1	LDP (SIMD&FP) - 64-bit variant on page C7-1356
10	0	0	STP - 64-bit variant on page C6-868
10	0	1	LDP - 64-bit variant on page C6-665
10	1	0	STP (SIMD&FP) - 128-bit variant on page C7-1629
10	1	1	LDP (SIMD&FP) - 128-bit variant on page C7-1356
11	-	-	Unallocated.

Load/store register pair (pre-indexed)

This section describes the encoding of the Load/store register pair (pre-indexed) instruction class. This section is decoded from [Loads and Stores on page C4-219](#).

31	30	29	28	27	26	25	24	23	22	21				15	14			10	9			5	4			0
opc	1	0	1	V	0	1	1	L						imm7				Rt2				Rn				Rt

Decode fields

Instruction page

opc	V	L	
00	0	0	STP - 32-bit variant on page C6-867
00	0	1	LDP - 32-bit variant on page C6-664
00	1	0	STP (SIMD&FP) - 32-bit variant on page C7-1628
00	1	1	LDP (SIMD&FP) - 32-bit variant on page C7-1355
01	0	0	Unallocated.
01	0	1	LDPSW

Decode fields			Instruction page
opc	V	L	
01	1	0	STP (SIMD&FP) - 64-bit variant on page C7-1628
01	1	1	LDP (SIMD&FP) - 64-bit variant on page C7-1355
10	0	0	STP - 64-bit variant on page C6-867
10	0	1	LDP - 64-bit variant on page C6-664
10	1	0	STP (SIMD&FP) - 128-bit variant on page C7-1628
10	1	1	LDP (SIMD&FP) - 128-bit variant on page C7-1355
11	-	-	Unallocated.

Load/store register (unscaled immediate)

This section describes the encoding of the Load/store register (unscaled immediate) instruction class. This section is decoded from *Loads and Stores* on page C4-219.

31	30	29	28	27	26	25	24	23	22	21	20					12	11	10	9			5	4			0
size	1	1	1	V	0	0	opc	0	imm9							0	0	Rn			Rt					

Decode fields			Instruction page
size	V	opc	
x1	1	1x	Unallocated.
00	0	00	STURB
00	0	01	LDURB
00	0	10	LDURSB - 64-bit variant on page C6-743
00	0	11	LDURSB - 32-bit variant on page C6-743
00	1	00	STUR (SIMD&FP) - 8-bit variant on page C7-1638
00	1	01	LDUR (SIMD&FP) - 8-bit variant on page C7-1367
00	1	10	STUR (SIMD&FP) - 128-bit variant on page C7-1638
00	1	11	LDUR (SIMD&FP) - 128-bit variant on page C7-1367
01	0	00	STURH
01	0	01	LDURH
01	0	10	LDURSH - 64-bit variant on page C6-745
01	0	11	LDURSH - 32-bit variant on page C6-745
01	1	00	STUR (SIMD&FP) - 16-bit variant on page C7-1638
01	1	01	LDUR (SIMD&FP) - 16-bit variant on page C7-1367
1x	0	11	Unallocated.

Decode fields			Instruction page
size	V	opc	
1x	1	1x	Unallocated.
10	0	00	STUR - 32-bit variant on page C6-917
10	0	01	LDUR - 32-bit variant on page C6-739
10	0	10	LDURSW
10	1	00	STUR (SIMD&FP) - 32-bit variant on page C7-1638
10	1	01	LDUR (SIMD&FP) - 32-bit variant on page C7-1367
11	0	00	STUR - 64-bit variant on page C6-917
11	0	01	LDUR - 64-bit variant on page C6-739
11	0	10	PRFM (unscaled offset)
11	1	00	STUR (SIMD&FP) - 64-bit variant on page C7-1638
11	1	01	LDUR (SIMD&FP) - 64-bit variant on page C7-1367

Load/store register (immediate post-indexed)

This section describes the encoding of the Load/store register (immediate post-indexed) instruction class. This section is decoded from [Loads and Stores on page C4-219](#).

31	30	29	28	27	26	25	24	23	22	21	20				12	11	10	9			5	4			0	
size	1	1	1	V	0	0	opc	0	imm9						0	1	Rn				Rt					

Decode fields			Instruction page
size	V	opc	
x1	1	1x	Unallocated.
00	0	00	STRB (immediate)
00	0	01	LDRB (immediate)
00	0	10	LDRSB (immediate) - 64-bit variant on page C6-685
00	0	11	LDRSB (immediate) - 32-bit variant on page C6-685
00	1	00	STR (immediate, SIMD&FP) - 8-bit variant on page C7-1631
00	1	01	LDR (immediate, SIMD&FP) - 8-bit variant on page C7-1358
00	1	10	STR (immediate, SIMD&FP) - 128-bit variant on page C7-1631
00	1	11	LDR (immediate, SIMD&FP) - 128-bit variant on page C7-1358
01	0	00	STRH (immediate)
01	0	01	LDRH (immediate)
01	0	10	LDRSH (immediate) - 64-bit variant on page C6-690

Decode fields			Instruction page
size	V	opc	
01	0	11	LDRSH (immediate) - 32-bit variant on page C6-690
01	1	00	STR (immediate, SIMD&FP) - 16-bit variant on page C7-1631
01	1	01	LDR (immediate, SIMD&FP) - 16-bit variant on page C7-1358
1x	0	11	Unallocated.
1x	1	1x	Unallocated.
10	0	00	STR (immediate) - 32-bit variant on page C6-870
10	0	01	LDR (immediate) - 32-bit variant on page C6-670
10	0	10	LDRSW (immediate)
10	1	00	STR (immediate, SIMD&FP) - 32-bit variant on page C7-1631
10	1	01	LDR (immediate, SIMD&FP) - 32-bit variant on page C7-1358
11	0	00	STR (immediate) - 64-bit variant on page C6-870
11	0	01	LDR (immediate) - 64-bit variant on page C6-670
11	0	10	Unallocated.
11	1	00	STR (immediate, SIMD&FP) - 64-bit variant on page C7-1631
11	1	01	LDR (immediate, SIMD&FP) - 64-bit variant on page C7-1358

Load/store register (unprivileged)

This section describes the encoding of the Load/store register (unprivileged) instruction class. This section is decoded from [Loads and Stores on page C4-219](#).

31	30	29	28	27	26	25	24	23	22	21	20					12	11	10	9			5	4			0
size	1	1	1	V	0	0	opc	0	imm9						1	0	Rn				Rt					

Decode fields			Instruction page
size	V	opc	
-	1	-	Unallocated.
00	0	00	STTRB
00	0	01	LDTRB
00	0	10	LDTRSB - 64-bit variant on page C6-722
00	0	11	LDTRSB - 32-bit variant on page C6-722
01	0	00	STTRH
01	0	01	LDTRH
01	0	10	LDTRSH - 64-bit variant on page C6-724

Decode fields			Instruction page
size	V	opc	
01	0	11	LDTRSH - 32-bit variant on page C6-724
1x	0	11	Unallocated.
10	0	00	STTR - 32-bit variant on page C6-901
10	0	01	LDTR - 32-bit variant on page C6-718
10	0	10	LDTRSW
11	0	00	STTR - 64-bit variant on page C6-901
11	0	01	LDTR - 64-bit variant on page C6-718
11	0	10	Unallocated.

Load/store register (immediate pre-indexed)

This section describes the encoding of the Load/store register (immediate pre-indexed) instruction class. This section is decoded from [Loads and Stores on page C4-219](#).

31	30	29	28	27	26	25	24	23	22	21	20					12	11	10	9			5	4					0
size	1	1	1	V	0	0	opc	0								imm9		1	1			Rn						Rt

Decode fields			Instruction page
size	V	opc	
x1	1	1x	Unallocated.
00	0	00	STRB (immediate)
00	0	01	LDRB (immediate)
00	0	10	LDRSB (immediate) - 64-bit variant on page C6-685
00	0	11	LDRSB (immediate) - 32-bit variant on page C6-685
00	1	00	STR (immediate, SIMD&FP) - 8-bit variant on page C7-1631
00	1	01	LDR (immediate, SIMD&FP) - 8-bit variant on page C7-1358
00	1	10	STR (immediate, SIMD&FP) - 128-bit variant on page C7-1632
00	1	11	LDR (immediate, SIMD&FP) - 128-bit variant on page C7-1359
01	0	00	STRH (immediate)
01	0	01	LDRH (immediate)
01	0	10	LDRSH (immediate) - 64-bit variant on page C6-690
01	0	11	LDRSH (immediate) - 32-bit variant on page C6-690
01	1	00	STR (immediate, SIMD&FP) - 16-bit variant on page C7-1632
01	1	01	LDR (immediate, SIMD&FP) - 16-bit variant on page C7-1359

Decode fields			Instruction page
size	V	opc	
1x	0	11	Unallocated.
1x	1	1x	Unallocated.
10	0	00	STR (immediate) - 32-bit variant on page C6-870
10	0	01	LDR (immediate) - 32-bit variant on page C6-670
10	0	10	LDRSW (immediate)
10	1	00	STR (immediate, SIMD&FP) - 32-bit variant on page C7-1632
10	1	01	LDR (immediate, SIMD&FP) - 32-bit variant on page C7-1359
11	0	00	STR (immediate) - 64-bit variant on page C6-870
11	0	01	LDR (immediate) - 64-bit variant on page C6-670
11	0	10	Unallocated.
11	1	00	STR (immediate, SIMD&FP) - 64-bit variant on page C7-1632
11	1	01	LDR (immediate, SIMD&FP) - 64-bit variant on page C7-1359

Atomic memory operations

This section describes the encoding of the Atomic memory operations instruction class. This section is decoded from *Loads and Stores* on page C4-219.

31	30	29	28	27	26	25	24	23	22	21	20	16	15	14	12	11	10	9	5	4	0
size	1	1	1	V	0	0	A	R	1	Rs			o3	opc	0	0	Rn			Rt	

Decode fields							Instruction page	Architecture version
size	V	A	R	o3	opc	Rt		
-	0	-	-	1	001	-	Unallocated.	-
-	0	-	-	1	01x	-	Unallocated.	-
-	0	-	-	1	101	-	Unallocated.	-
-	0	-	-	1	11x	-	Unallocated.	-
-	0	0	-	1	100	-	Unallocated.	-
-	0	1	1	1	100	-	Unallocated.	-
-	1	-	-	-	-	-	Unallocated.	-
00	0	0	0	0	000	!= 11111	LDADDB, LDADDAB, LDADDALB, LDADDLB - No memory ordering variant on page C6-632	ARMv8.1
00	0	0	0	0	000	11111	STADDB, STADDLB - No memory ordering variant on page C6-832	ARMv8.1

Decode fields							Instruction page	Architecture version
size	V	A	R	o3	opc	Rt		
00	0	0	0	0	001	!= 11111	LDCLRB, LDCLRAB, LDCLRALB, LDCLRLB - <i>No memory ordering variant on page C6-647</i>	ARMv8.1
00	0	0	0	0	001	11111	STCLRB, STCLRLB - <i>No memory ordering variant on page C6-838</i>	ARMv8.1
00	0	0	0	0	010	!= 11111	LDEORB, LDEORAB, LDEORALB, LDEORLB - <i>No memory ordering variant on page C6-653</i>	ARMv8.1
00	0	0	0	0	010	11111	STEORB, STEORLB - <i>No memory ordering variant on page C6-844</i>	ARMv8.1
00	0	0	0	0	011	!= 11111	LDSETB, LDSETAB, LDSETALB, LDSETLB - <i>No memory ordering variant on page C6-700</i>	ARMv8.1
00	0	0	0	0	011	11111	STSETB, STSETLB - <i>No memory ordering variant on page C6-883</i>	ARMv8.1
00	0	0	0	0	100	!= 11111	LDSMAXB, LDSMAXAB, LDSMAXALB, LDSMAXLB - <i>No memory ordering variant on page C6-706</i>	ARMv8.1
00	0	0	0	0	100	11111	STSMAXB, STSMAXLB - <i>No memory ordering variant on page C6-889</i>	ARMv8.1
00	0	0	0	0	101	!= 11111	LDSMINB, LDSMINAB, LDSMINALB, LDSMINLB - <i>No memory ordering variant on page C6-712</i>	ARMv8.1
00	0	0	0	0	101	11111	STSMINB, STSMINLB - <i>No memory ordering variant on page C6-895</i>	ARMv8.1
00	0	0	0	0	110	!= 11111	LDUMAXB, LDUMAXAB, LDUMAXALB, LDUMAXLB - <i>No memory ordering variant on page C6-727</i>	ARMv8.1
00	0	0	0	0	110	11111	STUMAXB, STUMAXLB - <i>No memory ordering variant on page C6-905</i>	ARMv8.1
00	0	0	0	0	111	!= 11111	LDUMINB, LDUMINAB, LDUMINALB, LDUMINLB - <i>No memory ordering variant on page C6-733</i>	ARMv8.1
00	0	0	0	0	111	11111	STUMINB, STUMINLB - <i>No memory ordering variant on page C6-911</i>	ARMv8.1
00	0	0	0	1	000	-	SWPB, SWPAB, SWPALB, SWPLB - <i>No memory ordering variant on page C6-941</i>	ARMv8.1
00	0	0	0	1	000	!= 11111	LDADDB, LDADDAB, LDADDALB, LDADDLB - <i>Release variant on page C6-632</i>	ARMv8.1
00	0	0	0	1	000	11111	STADDB, STADDLB - <i>Release variant on page C6-832</i>	ARMv8.1
00	0	0	0	1	001	!= 11111	LDCLRB, LDCLRAB, LDCLRALB, LDCLRLB - <i>Release variant on page C6-647</i>	ARMv8.1
00	0	0	0	1	001	11111	STCLRB, STCLRLB - <i>Release variant on page C6-838</i>	ARMv8.1

Decode fields							Instruction page	Architecture version
size	V	A	R	o3	opc	Rt		
00	0	0	1	0	010	!= 11111	LDEORB, LDEORAB, LDEORALB, LDEORLB - <i>Release variant on page C6-653</i>	ARMv8.1
00	0	0	1	0	010	11111	STEORB, STEORLB - <i>Release variant on page C6-844</i>	ARMv8.1
00	0	0	1	0	011	!= 11111	LDSETB, LDSETAB, LDSETALB, LDSETLB - <i>Release variant on page C6-700</i>	ARMv8.1
00	0	0	1	0	011	11111	STSETB, STSETLB - <i>Release variant on page C6-883</i>	ARMv8.1
00	0	0	1	0	100	!= 11111	LDSMAXB, LDSMAXAB, LDSMAXALB, LDSMAXLB - <i>Release variant on page C6-706</i>	ARMv8.1
00	0	0	1	0	100	11111	STSMAXB, STSMAXLB - <i>Release variant on page C6-889</i>	ARMv8.1
00	0	0	1	0	101	!= 11111	LDSMINB, LDSMINAB, LDSMINALB, LDSMINLB - <i>Release variant on page C6-712</i>	ARMv8.1
00	0	0	1	0	101	11111	STSMINB, STSMINLB - <i>Release variant on page C6-895</i>	ARMv8.1
00	0	0	1	0	110	!= 11111	LDUMAXB, LDUMAXAB, LDUMAXALB, LDUMAXLB - <i>Release variant on page C6-727</i>	ARMv8.1
00	0	0	1	0	110	11111	STUMAXB, STUMAXLB - <i>Release variant on page C6-905</i>	ARMv8.1
00	0	0	1	0	111	!= 11111	LDUMINB, LDUMINAB, LDUMINALB, LDUMINLB - <i>Release variant on page C6-733</i>	ARMv8.1
00	0	0	1	0	111	11111	STUMINB, STUMINLB - <i>Release variant on page C6-911</i>	ARMv8.1
00	0	0	1	1	000	-	SWPB, SWPAB, SWPALB, SWPLB - <i>Release variant on page C6-941</i>	ARMv8.1
00	0	1	0	0	000	-	LDADDB, LDADDAB, LDADDALB, LDADDLB - <i>Acquire variant on page C6-632</i>	ARMv8.1
00	0	1	0	0	001	-	LDCLRB, LDCLRAB, LDCLRALB, LDCLRLB - <i>Acquire variant on page C6-647</i>	ARMv8.1
00	0	1	0	0	010	-	LDEORB, LDEORAB, LDEORALB, LDEORLB - <i>Acquire variant on page C6-653</i>	ARMv8.1
00	0	1	0	0	011	-	LDSETB, LDSETAB, LDSETALB, LDSETLB - <i>Acquire variant on page C6-700</i>	ARMv8.1
00	0	1	0	0	100	-	LDSMAXB, LDSMAXAB, LDSMAXALB, LDSMAXLB - <i>Acquire variant on page C6-706</i>	ARMv8.1
00	0	1	0	0	101	-	LDSMINB, LDSMINAB, LDSMINALB, LDSMINLB - <i>Acquire variant on page C6-712</i>	ARMv8.1
00	0	1	0	0	110	-	LDUMAXB, LDUMAXAB, LDUMAXALB, LDUMAXLB - <i>Acquire variant on page C6-727</i>	ARMv8.1
00	0	1	0	0	111	-	LDUMINB, LDUMINAB, LDUMINALB, LDUMINLB - <i>Acquire variant on page C6-733</i>	ARMv8.1

Decode fields							Instruction page	Architecture version
size	V	A	R	o3	opc	Rt		
00	0	1	0	1	000	-	SWPB, SWPAB, SWPALB, SWPLB - <i>Acquire variant on page C6-941</i>	ARMv8.1
00	0	1	1	0	000	-	LDADDB, LDADDAB, LDADDALB, LDADDLB - <i>Acquire and release variant on page C6-632</i>	ARMv8.1
00	0	1	1	0	001	-	LDCLRB, LDCLRAB, LDCLRALB, LDCLRLB - <i>Acquire and release variant on page C6-647</i>	ARMv8.1
00	0	1	1	0	010	-	LDEORB, LDEORAB, LDEORALB, LDEORLB - <i>Acquire and release variant on page C6-653</i>	ARMv8.1
00	0	1	1	0	011	-	LDSETB, LDSETAB, LDSETALB, LDSETLB - <i>Acquire and release variant on page C6-700</i>	ARMv8.1
00	0	1	1	0	100	-	LDSMAXB, LDSMAXAB, LDSMAXALB, LDSMAXLB - <i>Acquire and release variant on page C6-706</i>	ARMv8.1
00	0	1	1	0	101	-	LDSMINB, LDSMINAB, LDSMINALB, LDSMINLB - <i>Acquire and release variant on page C6-712</i>	ARMv8.1
00	0	1	1	0	110	-	LDUMAXB, LDUMAXAB, LDUMAXALB, LDUMAXLB - <i>Acquire and release variant on page C6-727</i>	ARMv8.1
00	0	1	1	0	111	-	LDUMINB, LDUMINAB, LDUMINALB, LDUMINLB - <i>Acquire and release variant on page C6-733</i>	ARMv8.1
00	0	1	1	1	000	-	SWPB, SWPAB, SWPALB, SWPLB - <i>Acquire and release variant on page C6-941</i>	ARMv8.1
01	0	0	0	0	000	!= 11111	LDADDH, LDADDAH, LDADDALH, LDADDLH - <i>No memory ordering variant on page C6-634</i>	ARMv8.1
01	0	0	0	0	000	11111	STADDH, STADDLH - <i>No memory ordering variant on page C6-834</i>	ARMv8.1
01	0	0	0	0	001	!= 11111	LDCLRH, LDCLRAH, LDCLRALH, LDCLRLH - <i>No memory ordering variant on page C6-649</i>	ARMv8.1
01	0	0	0	0	001	11111	STCLRH, STCLRLH - <i>No memory ordering variant on page C6-840</i>	ARMv8.1
01	0	0	0	0	010	!= 11111	LDEORH, LDEORAH, LDEORALH, LDEORLH - <i>No memory ordering variant on page C6-655</i>	ARMv8.1
01	0	0	0	0	010	11111	STEORH, STEORLH - <i>No memory ordering variant on page C6-846</i>	ARMv8.1
01	0	0	0	0	011	!= 11111	LDSETH, LDSETHA, LDSETALH, LDSETLH - <i>No memory ordering variant on page C6-702</i>	ARMv8.1
01	0	0	0	0	011	11111	STSETH, STSETLH - <i>No memory ordering variant on page C6-885</i>	ARMv8.1

Decode fields							Instruction page	Architecture version
size	V	A	R	o3	opc	Rt		
01	0	0	0	0	100	!= 11111	LDSMAXH, LDSMAXAH, LDSMAXALH, LDSMAXLH - <i>No memory ordering variant on page C6-708</i>	ARMv8.1
01	0	0	0	0	100	11111	STSMAXH, STSMAXLH - <i>No memory ordering variant on page C6-891</i>	ARMv8.1
01	0	0	0	0	101	!= 11111	LDSMINH, LDSMINAH, LDSMINALH, LDSMINLH - <i>No memory ordering variant on page C6-714</i>	ARMv8.1
01	0	0	0	0	101	11111	STSMINH, STSMINLH - <i>No memory ordering variant on page C6-897</i>	ARMv8.1
01	0	0	0	0	110	!= 11111	LDUMAXH, LDUMAXAH, LDUMAXALH, LDUMAXLH - <i>No memory ordering variant on page C6-729</i>	ARMv8.1
01	0	0	0	0	110	11111	STUMAXH, STUMAXLH - <i>No memory ordering variant on page C6-907</i>	ARMv8.1
01	0	0	0	0	111	!= 11111	LDUMINH, LDUMINAH, LDUMINALH, LDUMINLH - <i>No memory ordering variant on page C6-735</i>	ARMv8.1
01	0	0	0	0	111	11111	STUMINH, STUMINLH - <i>No memory ordering variant on page C6-913</i>	ARMv8.1
01	0	0	0	1	000	-	SWPH, SWPAH, SWPALH, SWPLH - <i>No memory ordering variant on page C6-943</i>	ARMv8.1
01	0	0	1	0	000	!= 11111	LDADDH, LDADDAH, LDADDALH, LDADDLH - <i>Release variant on page C6-634</i>	ARMv8.1
01	0	0	1	0	000	11111	STADDH, STADDLH - <i>Release variant on page C6-834</i>	ARMv8.1
01	0	0	1	0	001	!= 11111	LDCLRH, LDCLRAH, LDCLRALH, LDCLRLH - <i>Release variant on page C6-649</i>	ARMv8.1
01	0	0	1	0	001	11111	STCLRH, STCLRLH - <i>Release variant on page C6-840</i>	ARMv8.1
01	0	0	1	0	010	!= 11111	LDEORH, LDEORAH, LDEORALH, LDEORLH - <i>Release variant on page C6-655</i>	ARMv8.1
01	0	0	1	0	010	11111	STEORH, STEORLH - <i>Release variant on page C6-846</i>	ARMv8.1
01	0	0	1	0	011	!= 11111	LDSETH, LDSETAH, LDSETALH, LDSETLH - <i>Release variant on page C6-702</i>	ARMv8.1
01	0	0	1	0	011	11111	STSETH, STSETLH - <i>Release variant on page C6-885</i>	ARMv8.1
01	0	0	1	0	100	!= 11111	LDSMAXH, LDSMAXAH, LDSMAXALH, LDSMAXLH - <i>Release variant on page C6-708</i>	ARMv8.1
01	0	0	1	0	100	11111	STSMAXH, STSMAXLH - <i>Release variant on page C6-891</i>	ARMv8.1
01	0	0	1	0	101	!= 11111	LDSMINH, LDSMINAH, LDSMINALH, LDSMINLH - <i>Release variant on page C6-714</i>	ARMv8.1

Decode fields							Instruction page	Architecture version
size	V	A	R	o3	opc	Rt		
01	0	0	1	0	101	11111	STSMINH, STSMINLH - <i>Release variant on page C6-897</i>	ARMv8.1
01	0	0	1	0	110	!= 11111	LDUMAXH, LDUMAXAH, LDUMAXALH, LDUMAXLH - <i>Release variant on page C6-729</i>	ARMv8.1
01	0	0	1	0	110	11111	STUMAXH, STUMAXLH - <i>Release variant on page C6-907</i>	ARMv8.1
01	0	0	1	0	111	!= 11111	LDUMINH, LDUMINAH, LDUMINALH, LDUMINLH - <i>Release variant on page C6-735</i>	ARMv8.1
01	0	0	1	0	111	11111	STUMINH, STUMINLH - <i>Release variant on page C6-913</i>	ARMv8.1
01	0	0	1	1	000	-	SWPH, SWPAH, SWPALH, SWPLH - <i>Release variant on page C6-943</i>	ARMv8.1
01	0	1	0	0	000	-	LDADDH, LDADDAH, LDADDALH, LDADDLH - <i>Acquire variant on page C6-634</i>	ARMv8.1
01	0	1	0	0	001	-	LDCLRH, LDCLRAH, LDCLRALH, LDCLRLH - <i>Acquire variant on page C6-649</i>	ARMv8.1
01	0	1	0	0	010	-	LDEORH, LDEORAH, LDEORALH, LDEORLH - <i>Acquire variant on page C6-655</i>	ARMv8.1
01	0	1	0	0	011	-	LDSETH, LDSETAH, LDSETALH, LDSETLH - <i>Acquire variant on page C6-702</i>	ARMv8.1
01	0	1	0	0	100	-	LDSMAXH, LDSMAXAH, LDSMAXALH, LDSMAXLH - <i>Acquire variant on page C6-708</i>	ARMv8.1
01	0	1	0	0	101	-	LDSMINH, LDSMINAH, LDSMINALH, LDSMINLH - <i>Acquire variant on page C6-714</i>	ARMv8.1
01	0	1	0	0	110	-	LDUMAXH, LDUMAXAH, LDUMAXALH, LDUMAXLH - <i>Acquire variant on page C6-729</i>	ARMv8.1
01	0	1	0	0	111	-	LDUMINH, LDUMINAH, LDUMINALH, LDUMINLH - <i>Acquire variant on page C6-735</i>	ARMv8.1
01	0	1	0	1	000	-	SWPH, SWPAH, SWPALH, SWPLH - <i>Acquire variant on page C6-943</i>	ARMv8.1
01	0	1	1	0	000	-	LDADDH, LDADDAH, LDADDALH, LDADDLH - <i>Acquire and release variant on page C6-634</i>	ARMv8.1
01	0	1	1	0	001	-	LDCLRH, LDCLRAH, LDCLRALH, LDCLRLH - <i>Acquire and release variant on page C6-649</i>	ARMv8.1
01	0	1	1	0	010	-	LDEORH, LDEORAH, LDEORALH, LDEORLH - <i>Acquire and release variant on page C6-655</i>	ARMv8.1
01	0	1	1	0	011	-	LDSETH, LDSETAH, LDSETALH, LDSETLH - <i>Acquire and release variant on page C6-702</i>	ARMv8.1
01	0	1	1	0	100	-	LDSMAXH, LDSMAXAH, LDSMAXALH, LDSMAXLH - <i>Acquire and release variant on page C6-708</i>	ARMv8.1

Decode fields							Instruction page	Architecture version
size	V	A	R	o3	opc	Rt		
01	0	1	1	0	101	-	LDSMINH, LDSMINAH, LDSMINALH, LDSMINLH - <i>Acquire and release variant on page C6-714</i>	ARMv8.1
01	0	1	1	0	110	-	LDUMAXH, LDUMAXAH, LDUMAXALH, LDUMAXLH - <i>Acquire and release variant on page C6-729</i>	ARMv8.1
01	0	1	1	0	111	-	LDUMINH, LDUMINAH, LDUMINALH, LDUMINLH - <i>Acquire and release variant on page C6-735</i>	ARMv8.1
01	0	1	1	1	000	-	SWPH, SWPAH, SWPALH, SWPLH - <i>Acquire and release variant on page C6-943</i>	ARMv8.1
10	0	0	0	0	000	!= 11111	LDADD, LDADDA, LDADDAL, LDADDL - <i>32-bit, no memory ordering variant on page C6-636</i>	ARMv8.1
10	0	0	0	0	000	11111	STADD, STADDL - <i>32-bit, no memory ordering variant on page C6-836</i>	ARMv8.1
10	0	0	0	0	001	!= 11111	LDCLR, LDCLRA, LDCLRAL, LDCLRL - <i>32-bit, no memory ordering variant on page C6-651</i>	ARMv8.1
10	0	0	0	0	001	11111	STCLR, STCLRL - <i>32-bit, no memory ordering variant on page C6-842</i>	ARMv8.1
10	0	0	0	0	010	!= 11111	LDEOR, LDEORA, LDEORAL, LDEORL - <i>32-bit, no memory ordering variant on page C6-657</i>	ARMv8.1
10	0	0	0	0	010	11111	STEOR, STEORL - <i>32-bit, no memory ordering variant on page C6-848</i>	ARMv8.1
10	0	0	0	0	011	!= 11111	LDSET, LDSETA, LDSETAL, LDSETL - <i>32-bit, no memory ordering variant on page C6-704</i>	ARMv8.1
10	0	0	0	0	011	11111	STSET, STSETL - <i>32-bit, no memory ordering variant on page C6-887</i>	ARMv8.1
10	0	0	0	0	100	!= 11111	LDSMAX, LDSMAXA, LDSMAXAL, LDSMAXL - <i>32-bit, no memory ordering variant on page C6-710</i>	ARMv8.1
10	0	0	0	0	100	11111	STSMAX, STSMAXL - <i>32-bit, no memory ordering variant on page C6-893</i>	ARMv8.1
10	0	0	0	0	101	!= 11111	LDSMIN, LDSMINA, LDSMINAL, LDSMINL - <i>32-bit, no memory ordering variant on page C6-716</i>	ARMv8.1
10	0	0	0	0	101	11111	STSMIN, STSMINL - <i>32-bit, no memory ordering variant on page C6-899</i>	ARMv8.1
10	0	0	0	0	110	!= 11111	LDUMAX, LDUMAXA, LDUMAXAL, LDUMAXL - <i>32-bit, no memory ordering variant on page C6-731</i>	ARMv8.1
10	0	0	0	0	110	11111	STUMAX, STUMAXL - <i>32-bit, no memory ordering variant on page C6-909</i>	ARMv8.1
10	0	0	0	0	111	!= 11111	LDUMIN, LDUMINA, LDUMINAL, LDUMINL - <i>32-bit, no memory ordering variant on page C6-737</i>	ARMv8.1

Decode fields							Instruction page	Architecture version
size	V	A	R	o3	opc	Rt		
10	0	0	0	0	111	11111	STUMIN, STUMINL - 32-bit, no memory ordering variant on page C6-915	ARMv8.1
10	0	0	0	1	000	-	SWP, SWPA, SWPAL, SWPL - 32-bit, no memory ordering variant on page C6-945	ARMv8.1
10	0	0	1	0	000	!= 11111	LDADD, LDADDA, LDADDAL, LDADDL - 32-bit, release variant on page C6-636	ARMv8.1
10	0	0	1	0	000	11111	STADD, STADDL - 32-bit, release variant on page C6-836	ARMv8.1
10	0	0	1	0	001	!= 11111	LDCLR, LDCLRA, LDCLRAL, LDCLRL - 32-bit, release variant on page C6-651	ARMv8.1
10	0	0	1	0	001	11111	STCLR, STCLRL - 32-bit, release variant on page C6-842	ARMv8.1
10	0	0	1	0	010	!= 11111	LDEOR, LDEORA, LDEORAL, LDEORL - 32-bit, release variant on page C6-657	ARMv8.1
10	0	0	1	0	010	11111	STEOR, STEORL - 32-bit, release variant on page C6-848	ARMv8.1
10	0	0	1	0	011	!= 11111	LDSET, LDSETA, LDSETAL, LDSETL - 32-bit, release variant on page C6-704	ARMv8.1
10	0	0	1	0	011	11111	STSET, STSETL - 32-bit, release variant on page C6-887	ARMv8.1
10	0	0	1	0	100	!= 11111	LDSMAX, LDSMAXA, LDSMAXAL, LDSMAXL - 32-bit, release variant on page C6-710	ARMv8.1
10	0	0	1	0	100	11111	STSMAX, STSMAXL - 32-bit, release variant on page C6-893	ARMv8.1
10	0	0	1	0	101	!= 11111	LDSMIN, LDSMINA, LDSMINAL, LDSMINL - 32-bit, release variant on page C6-716	ARMv8.1
10	0	0	1	0	101	11111	STSMIN, STSMINL - 32-bit, release variant on page C6-899	ARMv8.1
10	0	0	1	0	110	!= 11111	LDUMAX, LDUMAXA, LDUMAXAL, LDUMAXL - 32-bit, release variant on page C6-731	ARMv8.1
10	0	0	1	0	110	11111	STUMAX, STUMAXL - 32-bit, release variant on page C6-909	ARMv8.1
10	0	0	1	0	111	!= 11111	LDUMIN, LDUMINA, LDUMINAL, LDUMINL - 32-bit, release variant on page C6-737	ARMv8.1
10	0	0	1	0	111	11111	STUMIN, STUMINL - 32-bit, release variant on page C6-915	ARMv8.1
10	0	0	1	1	000	-	SWP, SWPA, SWPAL, SWPL - 32-bit, release variant on page C6-945	ARMv8.1
10	0	1	0	0	000	-	LDADD, LDADDA, LDADDAL, LDADDL - 32-bit, acquire variant on page C6-636	ARMv8.1

Decode fields							Instruction page	Architecture version
size	V	A	R	o3	opc	Rt		
10	0	1	0	0	001	-	LDCLR, LDCLRA, LDCLRAL, LDCLRL - 32-bit, acquire variant on page C6-651	ARMv8.1
10	0	1	0	0	010	-	LDEOR, LDEORA, LDEORAL, LDEORL - 32-bit, acquire variant on page C6-657	ARMv8.1
10	0	1	0	0	011	-	LDSET, LDSETA, LDSETAL, LDSETL - 32-bit, acquire variant on page C6-704	ARMv8.1
10	0	1	0	0	100	-	LDSMAX, LDSMAXA, LDSMAXAL, LDSMAXL - 32-bit, acquire variant on page C6-710	ARMv8.1
10	0	1	0	0	101	-	LDSMIN, LDSMINA, LDSMINAL, LDSMINL - 32-bit, acquire variant on page C6-716	ARMv8.1
10	0	1	0	0	110	-	LDUMAX, LDUMAXA, LDUMAXAL, LDUMAXL - 32-bit, acquire variant on page C6-731	ARMv8.1
10	0	1	0	0	111	-	LDUMIN, LDUMINA, LDUMINAL, LDUMINL - 32-bit, acquire variant on page C6-737	ARMv8.1
10	0	1	0	1	000	-	SWP, SWPA, SWPAL, SWPL - 32-bit, acquire variant on page C6-945	ARMv8.1
10	0	1	1	0	000	-	LDADD, LDADDA, LDADDAL, LDADDL - 32-bit, acquire and release variant on page C6-636	ARMv8.1
10	0	1	1	0	001	-	LDCLR, LDCLRA, LDCLRAL, LDCLRL - 32-bit, acquire and release variant on page C6-651	ARMv8.1
10	0	1	1	0	010	-	LDEOR, LDEORA, LDEORAL, LDEORL - 32-bit, acquire and release variant on page C6-657	ARMv8.1
10	0	1	1	0	011	-	LDSET, LDSETA, LDSETAL, LDSETL - 32-bit, acquire and release variant on page C6-704	ARMv8.1
10	0	1	1	0	100	-	LDSMAX, LDSMAXA, LDSMAXAL, LDSMAXL - 32-bit, acquire and release variant on page C6-710	ARMv8.1
10	0	1	1	0	101	-	LDSMIN, LDSMINA, LDSMINAL, LDSMINL - 32-bit, acquire and release variant on page C6-716	ARMv8.1
10	0	1	1	0	110	-	LDUMAX, LDUMAXA, LDUMAXAL, LDUMAXL - 32-bit, acquire and release variant on page C6-731	ARMv8.1
10	0	1	1	0	111	-	LDUMIN, LDUMINA, LDUMINAL, LDUMINL - 32-bit, acquire and release variant on page C6-737	ARMv8.1
10	0	1	1	1	000	-	SWP, SWPA, SWPAL, SWPL - 32-bit, acquire and release variant on page C6-945	ARMv8.1
11	0	0	0	0	000	!= 11111	LDADD, LDADDA, LDADDAL, LDADDL - 64-bit, no memory ordering variant on page C6-636	ARMv8.1
11	0	0	0	0	000	11111	STADD, STADDL - 64-bit, no memory ordering variant on page C6-836	ARMv8.1
11	0	0	0	0	001	!= 11111	LDCLR, LDCLRA, LDCLRAL, LDCLRL - 64-bit, no memory ordering variant on page C6-651	ARMv8.1

Decode fields							Instruction page	Architecture version
size	V	A	R	o3	opc	Rt		
11	0	0	0	0	001	11111	STCLR, STCLRL - 64-bit, no memory ordering variant on page C6-842	ARMv8.1
11	0	0	0	0	010	!= 11111	LDEOR, LDEORA, LDEORAL, LDEORL - 64-bit, no memory ordering variant on page C6-657	ARMv8.1
11	0	0	0	0	010	11111	STEOR, STEORL - 64-bit, no memory ordering variant on page C6-848	ARMv8.1
11	0	0	0	0	011	!= 11111	LDSET, LDSETA, LDSETAL, LDSETL - 64-bit, no memory ordering variant on page C6-704	ARMv8.1
11	0	0	0	0	011	11111	STSET, STSETL - 64-bit, no memory ordering variant on page C6-887	ARMv8.1
11	0	0	0	0	100	!= 11111	LDSMAX, LDSMAXA, LDSMAXAL, LDSMAXL - 64-bit, no memory ordering variant on page C6-710	ARMv8.1
11	0	0	0	0	100	11111	STSMAX, STSMAXL - 64-bit, no memory ordering variant on page C6-893	ARMv8.1
11	0	0	0	0	101	!= 11111	LDSMIN, LDSMINA, LDSMINAL, LDSMINL - 64-bit, no memory ordering variant on page C6-716	ARMv8.1
11	0	0	0	0	101	11111	STSMIN, STSMINL - 64-bit, no memory ordering variant on page C6-899	ARMv8.1
11	0	0	0	0	110	!= 11111	LDUMAX, LDUMAXA, LDUMAXAL, LDUMAXL - 64-bit, no memory ordering variant on page C6-731	ARMv8.1
11	0	0	0	0	110	11111	STUMAX, STUMAXL - 64-bit, no memory ordering variant on page C6-909	ARMv8.1
11	0	0	0	0	111	!= 11111	LDUMIN, LDUMINA, LDUMINAL, LDUMINL - 64-bit, no memory ordering variant on page C6-737	ARMv8.1
11	0	0	0	0	111	11111	STUMIN, STUMINL - 64-bit, no memory ordering variant on page C6-915	ARMv8.1
11	0	0	0	1	000	-	SWP, SWPA, SWPAL, SWPL - 64-bit, no memory ordering variant on page C6-945	ARMv8.1
11	0	0	1	0	000	!= 11111	LDADD, LDADDA, LDADDAL, LDADDL - 64-bit, release variant on page C6-637	ARMv8.1
11	0	0	1	0	000	11111	STADD, STADDL - 64-bit, release variant on page C6-836	ARMv8.1
11	0	0	1	0	001	!= 11111	LDCLR, LDCLRA, LDCLRAL, LDCLRL - 64-bit, release variant on page C6-652	ARMv8.1
11	0	0	1	0	001	11111	STCLR, STCLRL - 64-bit, release variant on page C6-842	ARMv8.1
11	0	0	1	0	010	!= 11111	LDEOR, LDEORA, LDEORAL, LDEORL - 64-bit, release variant on page C6-658	ARMv8.1
11	0	0	1	0	010	11111	STEOR, STEORL - 64-bit, release variant on page C6-848	ARMv8.1

Decode fields							Instruction page	Architecture version
size	V	A	R	o3	opc	Rt		
11	0	0	1	0	011	!= 11111	LDSET, LDSETA, LDSETAL, LDSETL - 64-bit, release variant on page C6-705	ARMv8.1
11	0	0	1	0	011	11111	STSET, STSETL - 64-bit, release variant on page C6-887	ARMv8.1
11	0	0	1	0	100	!= 11111	LDSMAX, LDSMAXA, LDSMAXAL, LDSMAXL - 64-bit, release variant on page C6-711	ARMv8.1
11	0	0	1	0	100	11111	STSMAX, STSMAXL - 64-bit, release variant on page C6-893	ARMv8.1
11	0	0	1	0	101	!= 11111	LDSMIN, LDSMINA, LDSMINAL, LDSMINL - 64-bit, release variant on page C6-717	ARMv8.1
11	0	0	1	0	101	11111	STSMIN, STSMINL - 64-bit, release variant on page C6-899	ARMv8.1
11	0	0	1	0	110	!= 11111	LDUMAX, LDUMAXA, LDUMAXAL, LDUMAXL - 64-bit, release variant on page C6-732	ARMv8.1
11	0	0	1	0	110	11111	STUMAX, STUMAXL - 64-bit, release variant on page C6-909	ARMv8.1
11	0	0	1	0	111	!= 11111	LDUMIN, LDUMINA, LDUMINAL, LDUMINL - 64-bit, release variant on page C6-738	ARMv8.1
11	0	0	1	0	111	11111	STUMIN, STUMINL - 64-bit, release variant on page C6-915	ARMv8.1
11	0	0	1	1	000	-	SWP, SWPA, SWPAL, SWPL - 64-bit, release variant on page C6-946	ARMv8.1
11	0	1	0	0	000	-	LDADD, LDADDA, LDADDAL, LDADDL - 64-bit, acquire variant on page C6-636	ARMv8.1
11	0	1	0	0	001	-	LDCLR, LDCLRA, LDCLRAL, LDCLRL - 64-bit, acquire variant on page C6-651	ARMv8.1
11	0	1	0	0	010	-	LDEOR, LDEORA, LDEORAL, LDEORL - 64-bit, acquire variant on page C6-657	ARMv8.1
11	0	1	0	0	011	-	LDSET, LDSETA, LDSETAL, LDSETL - 64-bit, acquire variant on page C6-704	ARMv8.1
11	0	1	0	0	100	-	LDSMAX, LDSMAXA, LDSMAXAL, LDSMAXL - 64-bit, acquire variant on page C6-710	ARMv8.1
11	0	1	0	0	101	-	LDSMIN, LDSMINA, LDSMINAL, LDSMINL - 64-bit, acquire variant on page C6-716	ARMv8.1
11	0	1	0	0	110	-	LDUMAX, LDUMAXA, LDUMAXAL, LDUMAXL - 64-bit, acquire variant on page C6-731	ARMv8.1
11	0	1	0	0	111	-	LDUMIN, LDUMINA, LDUMINAL, LDUMINL - 64-bit, acquire variant on page C6-737	ARMv8.1
11	0	1	0	1	000	-	SWP, SWPA, SWPAL, SWPL - 64-bit, acquire variant on page C6-945	ARMv8.1

Decode fields							Instruction page	Architecture version
size	V	A	R	o3	opc	Rt		
11	0	1	1	0	000	-	LDADD, LDADDA, LDADDAL, LDADDL - <i>64-bit, acquire and release variant on page C6-636</i>	ARMv8.1
11	0	1	1	0	001	-	LDCLR, LDCLRA, LDCLRAL, LDCLRL - <i>64-bit, acquire and release variant on page C6-651</i>	ARMv8.1
11	0	1	1	0	010	-	LDEOR, LDEORA, LDEORAL, LDEORL - <i>64-bit, acquire and release variant on page C6-657</i>	ARMv8.1
11	0	1	1	0	011	-	LDSET, LDSETA, LDSETAL, LDSETL - <i>64-bit, acquire and release variant on page C6-704</i>	ARMv8.1
11	0	1	1	0	100	-	LDSMAX, LDSMAXA, LDSMAXAL, LDSMAXL - <i>64-bit, acquire and release variant on page C6-710</i>	ARMv8.1
11	0	1	1	0	101	-	LDSMIN, LDSMINA, LDSMINAL, LDSMINL - <i>64-bit, acquire and release variant on page C6-716</i>	ARMv8.1
11	0	1	1	0	110	-	LDUMAX, LDUMAXA, LDUMAXAL, LDUMAXL - <i>64-bit, acquire and release variant on page C6-731</i>	ARMv8.1
11	0	1	1	0	111	-	LDUMIN, LDUMINA, LDUMINAL, LDUMINL - <i>64-bit, acquire and release variant on page C6-737</i>	ARMv8.1
11	0	1	1	1	000	-	SWP, SWPA, SWPAL, SWPL - <i>64-bit, acquire and release variant on page C6-945</i>	ARMv8.1

Load/store register (register offset)

This section describes the encoding of the Load/store register (register offset) instruction class. This section is decoded from *Loads and Stores on page C4-219*.

31	30	29	28	27	26	25	24	23	22	21	20	16	15	13	12	11	10	9	5	4	0
size	1	1	1	V	0	0	opc	1	Rm				option	S	1	0	Rn				Rt

Decode fields				Instruction page
size	V	opc	option	
-	-	-	x0x	Unallocated.
x1	1	1x	-	Unallocated.
00	0	00	!= 011	STRB (register) - <i>Extended register variant on page C6-877</i>
00	0	00	011	STRB (register) - <i>Shifted register variant on page C6-877</i>
00	0	01	!= 011	LDRB (register) - <i>Extended register variant on page C6-679</i>
00	0	01	011	LDRB (register) - <i>Shifted register variant on page C6-679</i>
00	0	10	!= 011	LDRSB (register) - <i>64-bit with extended register offset variant on page C6-688</i>
00	0	10	011	LDRSB (register) - <i>64-bit with shifted register offset variant on page C6-688</i>

Decode fields				Instruction page
size	V	opc	option	
00	0	11	!= 011	LDRSB (register) - 32-bit with extended register offset variant on page C6-688
00	0	11	011	LDRSB (register) - 32-bit with shifted register offset variant on page C6-688
00	1	00	!= 011	STR (register, SIMD&FP)
00	1	00	011	STR (register, SIMD&FP)
00	1	01	!= 011	LDR (register, SIMD&FP)
00	1	01	011	LDR (register, SIMD&FP)
00	1	10	-	STR (register, SIMD&FP)
00	1	11	-	LDR (register, SIMD&FP)
01	0	00	-	STRH (register)
01	0	01	-	LDRH (register)
01	0	10	-	LDRSH (register) - 64-bit variant on page C6-693
01	0	11	-	LDRSH (register) - 32-bit variant on page C6-693
01	1	00	-	STR (register, SIMD&FP)
01	1	01	-	LDR (register, SIMD&FP)
1x	0	11	-	Unallocated.
1x	1	1x	-	Unallocated.
10	0	00	-	STR (register) - 32-bit variant on page C6-873
10	0	01	-	LDR (register) - 32-bit variant on page C6-675
10	0	10	-	LDRSW (register)
10	1	00	-	STR (register, SIMD&FP)
10	1	01	-	LDR (register, SIMD&FP)
11	0	00	-	STR (register) - 64-bit variant on page C6-873
11	0	01	-	LDR (register) - 64-bit variant on page C6-675
11	0	10	-	PRFM (register)
11	1	00	-	STR (register, SIMD&FP)
11	1	01	-	LDR (register, SIMD&FP)

Load/store register (unsigned immediate)

This section describes the encoding of the Load/store register (unsigned immediate) instruction class. This section is decoded from [Loads and Stores](#) on page C4-219.

31	30	29	28	27	26	25	24	23	22	21					10	9			5	4			0
size	1	1	1	V	0	1	opc	imm12							Rn				Rt				

Decode fields			Instruction page
size	V	opc	
x1	1	1x	Unallocated.
00	0	00	STRB (immediate)
00	0	01	LDRB (immediate)
00	0	10	LDRSB (immediate) - 64-bit variant on page C6-686
00	0	11	LDRSB (immediate) - 32-bit variant on page C6-686
00	1	00	STR (immediate, SIMD&FP) - 8-bit variant on page C7-1632
00	1	01	LDR (immediate, SIMD&FP) - 8-bit variant on page C7-1359
00	1	10	STR (immediate, SIMD&FP) - 128-bit variant on page C7-1633
00	1	11	LDR (immediate, SIMD&FP) - 128-bit variant on page C7-1360
01	0	00	STRH (immediate)
01	0	01	LDRH (immediate)
01	0	10	LDRSH (immediate) - 64-bit variant on page C6-691
01	0	11	LDRSH (immediate) - 32-bit variant on page C6-691
01	1	00	STR (immediate, SIMD&FP) - 16-bit variant on page C7-1632
01	1	01	LDR (immediate, SIMD&FP) - 16-bit variant on page C7-1359
1x	0	11	Unallocated.
1x	1	1x	Unallocated.
10	0	00	STR (immediate) - 32-bit variant on page C6-871
10	0	01	LDR (immediate) - 32-bit variant on page C6-671
10	0	10	LDRSW (immediate)
10	1	00	STR (immediate, SIMD&FP) - 32-bit variant on page C7-1632
10	1	01	LDR (immediate, SIMD&FP) - 32-bit variant on page C7-1359
11	0	00	STR (immediate) - 64-bit variant on page C6-871
11	0	01	LDR (immediate) - 64-bit variant on page C6-671
11	0	10	PRFM (immediate)
11	1	00	STR (immediate, SIMD&FP) - 64-bit variant on page C7-1632
11	1	01	LDR (immediate, SIMD&FP) - 64-bit variant on page C7-1359

C4.1.4 Data Processing -- Register

This section describes the encoding of the Data Processing -- Register group. This section is decoded from *A64 instruction set encoding* on page C4-210.

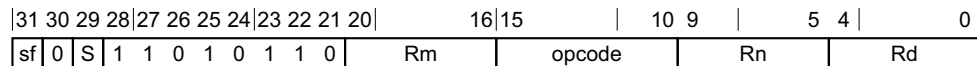


Table C4-5 Encoding table for the Data Processing -- Register group

Decode fields				Decode group or instruction page
op0	op1	op2	op3	
0	1	0110	-	Data-processing (2 source)
1	1	0110	-	Data-processing (1 source) on page C4-254
-	0	0xxx	-	Logical (shifted register) on page C4-255
-	0	1xx0	-	Add/subtract (shifted register) on page C4-256
-	0	1xx1	-	Add/subtract (extended register) on page C4-257
-	1	0000	-	Add/subtract (with carry) on page C4-258
-	1	0010	0	Conditional compare (register) on page C4-258
-	1	0010	1	Conditional compare (immediate) on page C4-259
-	1	0100	-	Conditional select on page C4-259
-	1	0xx1	-	Unallocated.
-	1	1xxx	-	Data-processing (3 source) on page C4-260

Data-processing (2 source)

This section describes the encoding of the Data-processing (2 source) instruction class. This section is decoded from *Data Processing -- Register*.



Decode fields			Instruction page
sf	S	opcode	
-	-	00000x	Unallocated.
-	-	011xxx	Unallocated.
-	-	1xxxxx	Unallocated.
-	0	0001xx	Unallocated.

Decode fields			Instruction page
sf	S	opcode	
-	0	0011xx	Unallocated.
-	1	-	Unallocated.
0	0	000010	UDIV - 32-bit variant on page C6-963
0	0	000011	SDIV - 32-bit variant on page C6-823
0	0	001000	LSLV - 32-bit variant on page C6-755
0	0	001001	LSRV - 32-bit variant on page C6-758
0	0	001010	ASRV - 32-bit variant on page C6-546
0	0	001011	RORV - 32-bit variant on page C6-814
0	0	010x11	Unallocated.
0	0	010000	CRC32B, CRC32H, CRC32W, CRC32X - CRC32B variant on page C6-595
0	0	010001	CRC32B, CRC32H, CRC32W, CRC32X - CRC32H variant on page C6-595
0	0	010010	CRC32B, CRC32H, CRC32W, CRC32X - CRC32W variant on page C6-595
0	0	010100	CRC32CB, CRC32CH, CRC32CW, CRC32CX - CRC32CB variant on page C6-597
0	0	010101	CRC32CB, CRC32CH, CRC32CW, CRC32CX - CRC32CH variant on page C6-597
0	0	010110	CRC32CB, CRC32CH, CRC32CW, CRC32CX - CRC32CW variant on page C6-597
1	0	000010	UDIV - 64-bit variant on page C6-963
1	0	000011	SDIV - 64-bit variant on page C6-823
1	0	001000	LSLV - 64-bit variant on page C6-755
1	0	001001	LSRV - 64-bit variant on page C6-758
1	0	001010	ASRV - 64-bit variant on page C6-546
1	0	001011	RORV - 64-bit variant on page C6-814
1	0	010xx0	Unallocated.
1	0	010x0x	Unallocated.
1	0	010011	CRC32B, CRC32H, CRC32W, CRC32X - CRC32X variant on page C6-595
1	0	010111	CRC32CB, CRC32CH, CRC32CW, CRC32CX - CRC32CX variant on page C6-597

Data-processing (1 source)

This section describes the encoding of the Data-processing (1 source) instruction class. This section is decoded from [Data Processing -- Register](#) on page C4-253.

31	30	29	28	27	26	25	24	23	22	21	20	16	15	10	9	5	4	0
sf	1	S	1	1	0	1	0	1	1	0	opcode2	opcode	Rn	Rd				

Decode fields				Instruction page	
sf	S	opcode2	opcode		
-	-	-	xx1xxx	Unallocated.	
-	-	-	x1xxxx	Unallocated.	
-	-	-	1xxxxx	Unallocated.	
-	-	xxx1	-	Unallocated.	
-	-	xxx1x	-	Unallocated.	
-	-	xx1xx	-	Unallocated.	
-	-	x1xxx	-	Unallocated.	
-	-	1xxxx	-	Unallocated.	
-	0	00000	00011x	Unallocated.	
-	1	-	-	Unallocated.	
0	0	00000	000000	RBIT - 32-bit variant on page C6-804	
0	0	00000	000001	REV16 - 32-bit variant on page C6-808	
0	0	00000	000010	REV - 32-bit variant on page C6-806	
0	0	00000	000011	Unallocated.	
0	0	00000	000100	CLZ - 32-bit variant on page C6-583	
0	0	00000	000101	CLS - 32-bit variant on page C6-582	
1	0	00000	000000	RBIT - 64-bit variant on page C6-804	
1	0	00000	000001	REV16 - 64-bit variant on page C6-808	
1	0	00000	000010	REV32	
1	0	00000	000011	REV - 64-bit variant on page C6-806	
1	0	00000	000100	CLZ - 64-bit variant on page C6-583	
1	0	00000	000101	CLS - 64-bit variant on page C6-582	

Logical (shifted register)

This section describes the encoding of the Logical (shifted register) instruction class. This section is decoded from [Data Processing -- Register on page C4-253](#).

31	30	29	28	27	26	25	24	23	22	21	20	16	15	10	9	5	4	0
sf	opc	0	1	0	1	0	shift	N	Rm				imm6		Rn		Rd	

Decode fields				Instruction page
sf	opc	N	imm6	
0	-	-	1xxxxx	Unallocated.
0	00	0	-	AND (shifted register) - 32-bit variant on page C6-538
0	00	1	-	BIC (shifted register) - 32-bit variant on page C6-556
0	01	0	-	ORR (shifted register) - 32-bit variant on page C6-792
0	01	1	-	ORN (shifted register) - 32-bit variant on page C6-788
0	10	0	-	EOR (shifted register) - 32-bit variant on page C6-620
0	10	1	-	EON (shifted register) - 32-bit variant on page C6-617
0	11	0	-	ANDS (shifted register) - 32-bit variant on page C6-542
0	11	1	-	BICS (shifted register) - 32-bit variant on page C6-558
1	00	0	-	AND (shifted register) - 64-bit variant on page C6-538
1	00	1	-	BIC (shifted register) - 64-bit variant on page C6-556
1	01	0	-	ORR (shifted register) - 64-bit variant on page C6-792
1	01	1	-	ORN (shifted register) - 64-bit variant on page C6-788
1	10	0	-	EOR (shifted register) - 64-bit variant on page C6-620
1	10	1	-	EON (shifted register) - 64-bit variant on page C6-617
1	11	0	-	ANDS (shifted register) - 64-bit variant on page C6-542
1	11	1	-	BICS (shifted register) - 64-bit variant on page C6-558

Add/subtract (shifted register)

This section describes the encoding of the Add/subtract (shifted register) instruction class. This section is decoded from [Data Processing -- Register on page C4-253](#).

31	30	29	28	27	26	25	24	23	22	21	20	16			15	10		9	5		4	0	
sf	op	S	0	1	0	1	1	shift	0	Rm			imm6			Rn			Rd				

Decode fields					Instruction page
sf	op	S	shift	imm6	
-	-	-	11	-	Unallocated.
0	-	-	-	1xxxxx	Unallocated.
0	0	0	-	-	ADD (shifted register) - 32-bit variant on page C6-527
0	0	1	-	-	ADDS (shifted register) - 32-bit variant on page C6-533
0	1	0	-	-	SUB (shifted register) - 32-bit variant on page C6-932
0	1	1	-	-	SUBS (shifted register) - 32-bit variant on page C6-938
1	0	0	-	-	ADD (shifted register) - 64-bit variant on page C6-527
1	0	1	-	-	ADDS (shifted register) - 64-bit variant on page C6-533
1	1	0	-	-	SUB (shifted register) - 64-bit variant on page C6-932
1	1	1	-	-	SUBS (shifted register) - 64-bit variant on page C6-938

Add/subtract (extended register)

This section describes the encoding of the Add/subtract (extended register) instruction class. This section is decoded from [Data Processing -- Register](#) on page C4-253.

31	30	29	28	27	26	25	24	23	22	21	20	16		15	13	12	10		9	5		4	0	
sf	op	S	0	1	0	1	1	opt	1	Rm			option			imm3			Rn			Rd		

Decode fields					Instruction page
sf	op	S	opt	imm3	
-	-	-	-	1x1	Unallocated.
-	-	-	-	11x	Unallocated.
-	-	-	x1	-	Unallocated.
-	-	-	1x	-	Unallocated.
0	0	0	00	-	ADD (extended register) - 32-bit variant on page C6-523
0	0	1	00	-	ADDS (extended register) - 32-bit variant on page C6-529
0	1	0	00	-	SUB (extended register) - 32-bit variant on page C6-928
0	1	1	00	-	SUBS (extended register) - 32-bit variant on page C6-934
1	0	0	00	-	ADD (extended register) - 64-bit variant on page C6-523

Decode fields					Instruction page
sf	op	S	opt	imm3	
1	0	1	00	-	ADDS (extended register) - 64-bit variant on page C6-529
1	1	0	00	-	SUB (extended register) - 64-bit variant on page C6-928
1	1	1	00	-	SUBS (extended register) - 64-bit variant on page C6-934

Add/subtract (with carry)

This section describes the encoding of the Add/subtract (with carry) instruction class. This section is decoded from [Data Processing -- Register on page C4-253](#).

31	30	29	28	27	26	25	24	23	22	21	20	16	15	10	9	5	4	0
sf	op	S	1	1	0	1	0	0	0	0	0	Rm	opcode2	Rn				Rd

Decode fields				Instruction page
sf	op	S	opcode2	
-	-	-	xxxxx1	Unallocated.
-	-	-	xxxx1x	Unallocated.
-	-	-	xxx1xx	Unallocated.
-	-	-	xx1xxx	Unallocated.
-	-	-	x1xxxx	Unallocated.
-	-	-	1xxxxx	Unallocated.
0	0	0	000000	ADC - 32-bit variant on page C6-521
0	0	1	000000	ADCS - 32-bit variant on page C6-522
0	1	0	000000	SBC - 32-bit variant on page C6-815
0	1	1	000000	SBCS - 32-bit variant on page C6-817
1	0	0	000000	ADC - 64-bit variant on page C6-521
1	0	1	000000	ADCS - 64-bit variant on page C6-522
1	1	0	000000	SBC - 64-bit variant on page C6-815
1	1	1	000000	SBCS - 64-bit variant on page C6-817

Conditional compare (register)

This section describes the encoding of the Conditional compare (register) instruction class. This section is decoded from [Data Processing -- Register on page C4-253](#).

31	30	29	28	27	26	25	24	23	22	21	20	16	15	12	11	10	9	5	4	3	0
sf	op	S	1	1	0	1	0	0	1	0		Rm		cond	0	o2		Rn	o3		nzcv

Decode fields

Instruction page

sf	op	S	o2	o3	
-	-	-	-	1	Unallocated.
-	-	-	1	-	Unallocated.
-	-	0	-	-	Unallocated.
0	0	1	0	0	CCMN (register) - 32-bit variant on page C6-576
0	1	1	0	0	CCMP (register) - 32-bit variant on page C6-578
1	0	1	0	0	CCMN (register) - 64-bit variant on page C6-576
1	1	1	0	0	CCMP (register) - 64-bit variant on page C6-578

Conditional compare (immediate)

This section describes the encoding of the Conditional compare (immediate) instruction class. This section is decoded from [Data Processing -- Register](#) on page C4-253.

31	30	29	28	27	26	25	24	23	22	21	20	16	15	12	11	10	9	5	4	3	0
sf	op	S	1	1	0	1	0	0	1	0		imm5		cond	1	o2		Rn	o3		nzcv

Decode fields

Instruction page

sf	op	S	o2	o3	
-	-	-	-	1	Unallocated.
-	-	-	1	-	Unallocated.
-	-	0	-	-	Unallocated.
0	0	1	0	0	CCMN (immediate) - 32-bit variant on page C6-575
0	1	1	0	0	CCMP (immediate) - 32-bit variant on page C6-577
1	0	1	0	0	CCMN (immediate) - 64-bit variant on page C6-575
1	1	1	0	0	CCMP (immediate) - 64-bit variant on page C6-577

Conditional select

This section describes the encoding of the Conditional select instruction class. This section is decoded from [Data Processing -- Register](#) on page C4-253.

31	30	29	28	27	26	25	24	23	22	21	20	16	15	12	11	10	9	5	4	0
sf	op	S	1	1	0	1	0	1	0	0		Rm		cond		op2		Rn		Rd

Decode fields				Instruction page
sf	op	S	op2	
-	-	-	1x	Unallocated.
-	-	1	-	Unallocated.
0	0	0	00	CSEL - 32-bit variant on page C6-599
0	0	0	01	CSINC - 32-bit variant on page C6-602
0	1	0	00	CSINV - 32-bit variant on page C6-604
0	1	0	01	CSNEG - 32-bit variant on page C6-606
1	0	0	00	CSEL - 64-bit variant on page C6-599
1	0	0	01	CSINC - 64-bit variant on page C6-602
1	1	0	00	CSINV - 64-bit variant on page C6-604
1	1	0	01	CSNEG - 64-bit variant on page C6-606

Data-processing (3 source)

This section describes the encoding of the Data-processing (3 source) instruction class. This section is decoded from [Data Processing -- Register on page C4-253](#).

31	30	29	28	27	26	25	24	23	21	20	16	15	14	10	9	5	4	0
sf	op54	1	1	0	1	1		op31			Rm	o0		Ra		Rn		Rd

Decode fields				Instruction page
sf	op54	op31	o0	
-	00	010	1	Unallocated.
-	00	011	-	Unallocated.
-	00	100	-	Unallocated.
-	00	110	1	Unallocated.
-	00	111	-	Unallocated.
-	01	-	-	Unallocated.
-	1x	-	-	Unallocated.
0	00	000	0	MADD - 32-bit variant on page C6-759
0	00	000	1	MSUB - 32-bit variant on page C6-776

Decode fields				Instruction page
sf	op54	op31	o0	
0	00	001	0	Unallocated.
0	00	001	1	Unallocated.
0	00	010	0	Unallocated.
0	00	101	0	Unallocated.
0	00	101	1	Unallocated.
0	00	110	0	Unallocated.
1	00	000	0	MADD - 64-bit variant on page C6-759
1	00	000	1	MSUB - 64-bit variant on page C6-776
1	00	001	0	SMADDL
1	00	001	1	SMSUBL
1	00	010	0	SMULH
1	00	101	0	UMADDL
1	00	101	1	UMSUBL
1	00	110	0	UMULH

C4.1.5 Data Processing -- Scalar Floating-Point and Advanced SIMD

This section describes the encoding of the Data Processing -- Scalar Floating-Point and Advanced SIMD group. This section is decoded from [A64 instruction set encoding on page C4-210](#).

31	28 27	24 23 22	19 18 17 16 15	10 9	0
op0	111	op1	op2	op3	op4

Table C4-6 Encoding table for the Data Processing -- Scalar Floating-Point and Advanced SIMD group

Decode fields					Decode group or instruction page
op0	op1	op2	op3	op4	
0000	0x	x101	00	xxxx10	Unallocated.
0010	0x	x101	00	xxxx10	Unallocated.
0100	0x	x101	00	xxxx10	Cryptographic AES on page C4-263
0101	0x	x0xx	-	0xxx00	Cryptographic three-register SHA on page C4-264
0101	0x	x0xx	-	0xxx10	Unallocated.
0101	0x	x101	00	xxxx10	Cryptographic two-register SHA on page C4-265
0110	0x	x101	00	xxxx10	Unallocated.
0111	0x	x0xx	-	0xxxx0	Unallocated.

Table C4-6 Encoding table for the Data Processing -- Scalar Floating-Point and Advanced SIMD group (continued)

Decode fields					Decode group or instruction page
op0	op1	op2	op3	op4	
0111	0x	x101	00	xxxx10	Unallocated.
01x1	00	00xx	-	0xxxx1	Advanced SIMD scalar copy on page C4-265
01x1	01	00xx	-	0xxxx1	Unallocated.
01x1	0x	0111	00	xxxx10	Unallocated.
01x1	0x	10xx	-	00xxx1	Advanced SIMD scalar three same FP16 on page C4-266
01x1	0x	10xx	-	01xxx1	Unallocated.
01x1	0x	1111	00	xxxx10	Advanced SIMD scalar two-register miscellaneous FP16 on page C4-266
01x1	0x	x0xx	-	1xxxx0	Unallocated.
01x1	0x	x0xx	-	1xxxx1	Advanced SIMD scalar three same extra on page C4-268
01x1	0x	x100	00	xxxx10	Advanced SIMD scalar two-register miscellaneous on page C4-268
01x1	0x	x110	00	xxxx10	Advanced SIMD scalar pairwise on page C4-270
01x1	0x	x1xx	1x	xxxx10	Unallocated.
01x1	0x	x1xx	x1	xxxx10	Unallocated.
01x1	0x	x1xx	-	xxxx00	Advanced SIMD scalar three different on page C4-271
01x1	0x	x1xx	-	xxxxx1	Advanced SIMD scalar three same on page C4-272
01x1	10	-	-	xxxxx1	Advanced SIMD scalar shift by immediate on page C4-274
01x1	11	-	-	xxxxx1	Unallocated.
01x1	1x	-	-	xxxxx0	Advanced SIMD scalar x indexed element on page C4-276
0x00	0x	x0xx	-	0xxx00	Advanced SIMD table lookup on page C4-277
0x00	0x	x0xx	-	0xxx10	Advanced SIMD permute on page C4-278
0x10	0x	x0xx	-	0xxxx0	Advanced SIMD extract on page C4-279
0xx0	00	00xx	-	0xxxx1	Advanced SIMD copy on page C4-279
0xx0	01	00xx	-	0xxxx1	Unallocated.
0xx0	0x	0111	00	xxxx10	Unallocated.
0xx0	0x	10xx	-	00xxx1	Advanced SIMD three same (FP16) on page C4-280
0xx0	0x	10xx	-	01xxx1	Unallocated.
0xx0	0x	1111	00	xxxx10	Advanced SIMD two-register miscellaneous (FP16) on page C4-281
0xx0	0x	x0xx	-	1xxxx0	Unallocated.
0xx0	0x	x0xx	-	1xxxx1	Advanced SIMD three same extra on page C4-282
0xx0	0x	x100	00	xxxx10	Advanced SIMD two-register miscellaneous on page C4-283
0xx0	0x	x110	00	xxxx10	Advanced SIMD across lanes on page C4-286

Table C4-6 Encoding table for the Data Processing -- Scalar Floating-Point and Advanced SIMD group (continued)

Decode fields					Decode group or instruction page
op0	op1	op2	op3	op4	
0xx0	0x	x1xx	1x	xxxx10	Unallocated.
0xx0	0x	x1xx	x1	xxxx10	Unallocated.
0xx0	0x	x1xx	-	xxxx00	Advanced SIMD three different on page C4-287
0xx0	0x	x1xx	-	xxxxx1	Advanced SIMD three same on page C4-288
0xx0	10	0000	-	xxxxx1	Advanced SIMD modified immediate on page C4-291
0xx0	10	!= 0000	-	xxxxx1	Advanced SIMD shift by immediate on page C4-292
0xx0	11	-	-	xxxxx1	Unallocated.
0xx0	1x	-	-	xxxxx0	Advanced SIMD vector x indexed element on page C4-294
11x1	-	-	-	-	Unallocated.
1xx0	-	-	-	-	Unallocated.
x0x1	0x	x0xx	-	-	Conversion between floating-point and fixed-point on page C4-295
x0x1	0x	x1xx	-	000000	Conversion between floating-point and integer on page C4-297
x0x1	0x	x1xx	-	100000	Unallocated.
x0x1	0x	x1xx	-	x10000	Floating-point data-processing (1 source) on page C4-302
x0x1	0x	x1xx	-	xx1000	Floating-point compare on page C4-304
x0x1	0x	x1xx	-	xxx100	Floating-point immediate on page C4-305
x0x1	0x	x1xx	-	xxxx01	Floating-point conditional compare on page C4-306
x0x1	0x	x1xx	-	xxxx10	Floating-point data-processing (2 source) on page C4-306
x0x1	0x	x1xx	-	xxxx11	Floating-point conditional select on page C4-307
x0x1	1x	-	-	-	Floating-point data-processing (3 source) on page C4-308

Cryptographic AES

This section describes the encoding of the Cryptographic AES instruction class. This section is decoded from [Data Processing -- Scalar Floating-Point and Advanced SIMD](#) on page C4-261.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	12	11	10	9	5	4	0
0	1	0	0	1	1	1	0	size	1	0	1	0	0	opcode	1	0	Rn					Rd

Decode fields		Instruction page
size	opcode	
-	x1xxx	Unallocated.
-	000xx	Unallocated.
-	1xxxx	Unallocated.
x1	-	Unallocated.
00	00100	AESE
00	00101	AESD
00	00110	AESMC
00	00111	AESIMC
1x	-	Unallocated.

Cryptographic three-register SHA

This section describes the encoding of the Cryptographic three-register SHA instruction class. This section is decoded from [Data Processing -- Scalar Floating-Point and Advanced SIMD on page C4-261](#).

31	30	29	28	27	26	25	24	23	22	21	20	16	15	14	12	11	10	9	5	4	0
0	1	0	1	1	1	1	0	size	0	Rm	0	opcode	0	0	Rn						Rd

Decode fields		Instruction page
size	opcode	
-	111	Unallocated.
x1	-	Unallocated.
00	000	SHA1C
00	001	SHA1P
00	010	SHA1M
00	011	SHA1SU0
00	100	SHA256H
00	101	SHA256H2
00	110	SHA256SU1
1x	-	Unallocated.

Cryptographic two-register SHA

This section describes the encoding of the Cryptographic two-register SHA instruction class. This section is decoded from *Data Processing -- Scalar Floating-Point and Advanced SIMD* on page C4-261.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	12	11	10	9	5	4	0
0	1	0	1	1	1	1	0	size	1	0	1	0	0	opcode	1	0	Rn					Rd

Decode fields		Instruction page
size	opcode	
-	xx1xx	Unallocated.
-	x1xxx	Unallocated.
-	1xxxx	Unallocated.
x1	-	Unallocated.
00	00000	SHA1H
00	00001	SHA1SU1
00	00010	SHA256SU0
00	00011	Unallocated.
1x	-	Unallocated.

Advanced SIMD scalar copy

This section describes the encoding of the Advanced SIMD scalar copy instruction class. This section is decoded from *Data Processing -- Scalar Floating-Point and Advanced SIMD* on page C4-261.

31	30	29	28	27	26	25	24	23	22	21	20	16	15	14	11	10	9	5	4	0
0	1	op	1	1	1	1	0	0	0	0	0	imm5	0	imm4	1	Rn				Rd

Decode fields			Instruction page
op	imm5	imm4	
0	-	xxx1	Unallocated.
0	-	xx1x	Unallocated.
0	-	x1xx	Unallocated.
0	-	0000	DUP (element)
0	-	1xxx	Unallocated.
0	x0000	0000	Unallocated.
1	-	-	Unallocated.

Advanced SIMD scalar three same FP16

This section describes the encoding of the Advanced SIMD scalar three same FP16 instruction class. This section is decoded from *Data Processing -- Scalar Floating-Point and Advanced SIMD* on page C4-261.

31	30	29	28	27	26	25	24	23	22	21	20	16	15	14	13	11	10	9	5	4	0
0	1	U	1	1	1	1	0	a	1	0	Rm	0	0	opcode	1	Rn					Rd

Decode fields			Instruction page	Architecture version
U	a	opcode		
-	-	110	Unallocated.	-
-	1	011	Unallocated.	-
0	0	011	FMULX	ARMv8.2
0	0	100	FCMEQ (register)	ARMv8.2
0	0	101	Unallocated.	-
0	0	111	FRECPS	ARMv8.2
0	1	100	Unallocated.	-
0	1	101	Unallocated.	-
0	1	111	FRSQRTS	ARMv8.2
1	0	011	Unallocated.	-
1	0	100	FCMGE (register)	ARMv8.2
1	0	101	FACGE	ARMv8.2
1	0	111	Unallocated.	-
1	1	010	FABD	ARMv8.2
1	1	100	FCMGT (register)	ARMv8.2
1	1	101	FACGT	ARMv8.2
1	1	111	Unallocated.	-

Advanced SIMD scalar two-register miscellaneous FP16

This section describes the encoding of the Advanced SIMD scalar two-register miscellaneous FP16 instruction class. This section is decoded from *Data Processing -- Scalar Floating-Point and Advanced SIMD* on page C4-261.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	12	11	10	9	5	4	0
0	1	U	1	1	1	1	0	a	1	1	1	1	0	0	opcode	1	0	Rn			Rd	

Decode fields			Instruction page	Architecture version
U	a	opcode		
-	-	00xxx	Unallocated.	-
-	-	010xx	Unallocated.	-
-	-	10xxx	Unallocated.	-
-	-	1100x	Unallocated.	-
-	-	11110	Unallocated.	-
-	0	011xx	Unallocated.	-
-	0	11111	Unallocated.	-
-	1	01111	Unallocated.	-
-	1	11100	Unallocated.	-
0	0	11010	FCVTNS (vector)	ARMv8.2
0	0	11011	FCVTMS (vector)	ARMv8.2
0	0	11100	FCVTAS (vector)	ARMv8.2
0	0	11101	SCVTF (vector, integer)	ARMv8.2
0	1	01100	FCMGT (zero)	ARMv8.2
0	1	01101	FCMEQ (zero)	ARMv8.2
0	1	01110	FCMLT (zero)	ARMv8.2
0	1	11010	FCVTPS (vector)	ARMv8.2
0	1	11011	FCVTZS (vector, integer)	ARMv8.2
0	1	11101	FRECPE	ARMv8.2
0	1	11111	FRECPX	ARMv8.2
1	0	11010	FCVTNU (vector)	ARMv8.2
1	0	11011	FCVTMU (vector)	ARMv8.2
1	0	11100	FCVTAU (vector)	ARMv8.2
1	0	11101	UCVTF (vector, integer)	ARMv8.2
1	1	01100	FCMGE (zero)	ARMv8.2
1	1	01101	FCMLE (zero)	ARMv8.2
1	1	01110	Unallocated.	-
1	1	11010	FCVTPU (vector)	ARMv8.2

Decode fields			Instruction page	Architecture version
U	a	opcode		
1	1	11011	FCVTZU (vector, integer)	ARMv8.2
1	1	11101	FRSQRT	ARMv8.2
1	1	11111	Unallocated.	-

Advanced SIMD scalar three same extra

This section describes the encoding of the Advanced SIMD scalar three same extra instruction class. This section is decoded from *Data Processing -- Scalar Floating-Point and Advanced SIMD* on page C4-261.

31	30	29	28	27	26	25	24	23	22	21	20	16			15	14	11			10	9	5			4	0		
0	1	U	1	1	1	1	0	size	0	Rm			1	opcode			1	Rn			Rd							

Decode fields		Instruction page	Architecture version
U	opcode		
-	001x	Unallocated.	-
-	01xx	Unallocated.	-
-	1xxx	Unallocated.	-
0	0000	Unallocated.	-
0	0001	Unallocated.	-
1	0000	SQRDMLAH (vector)	ARMv8.1
1	0001	SQRDMLSH (vector)	ARMv8.1

Advanced SIMD scalar two-register miscellaneous

This section describes the encoding of the Advanced SIMD scalar two-register miscellaneous instruction class. This section is decoded from *Data Processing -- Scalar Floating-Point and Advanced SIMD* on page C4-261.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	12				11	10	9	5		4	0	
0	1	U	1	1	1	1	0	size	1	0	0	0	0	opcode			1	0	Rn			Rd					

Decode fields			Instruction page
U	size	opcode	
-	-	0000x	Unallocated.
-	-	00010	Unallocated.
-	-	0010x	Unallocated.
-	-	00110	Unallocated.

Decode fields			Instruction page
U	size	opcode	
-	-	01111	Unallocated.
-	-	1000x	Unallocated.
-	-	10011	Unallocated.
-	-	10101	Unallocated.
-	-	10111	Unallocated.
-	-	1100x	Unallocated.
-	-	11110	Unallocated.
-	0x	011xx	Unallocated.
-	0x	11111	Unallocated.
-	1x	10110	Unallocated.
-	1x	11100	Unallocated.
0	-	00011	SUQADD
0	-	00111	SQABS
0	-	01000	CMGT (zero)
0	-	01001	CMEQ (zero)
0	-	01010	CMLT (zero)
0	-	01011	ABS
0	-	10010	Unallocated.
0	-	10100	SQXTN, SQXTN2
0	0x	10110	Unallocated.
0	0x	11010	FCVTNS (vector)
0	0x	11011	FCVTMS (vector)
0	0x	11100	FCVTAS (vector)
0	0x	11101	SCVTF (vector, integer)
0	1x	01100	FCMGT (zero)
0	1x	01101	FCMEQ (zero)
0	1x	01110	FCMLT (zero)
0	1x	11010	FCVTPS (vector)
0	1x	11011	FCVTZS (vector, integer)
0	1x	11101	FRECPE
0	1x	11111	FRECPX
1	-	00011	USQADD

Decode fields			Instruction page
U	size	opcode	
1	-	00111	SQNEG
1	-	01000	CMGE (zero)
1	-	01001	CMLE (zero)
1	-	01010	Unallocated.
1	-	01011	NEG (vector)
1	-	10010	SQXTUN, SQXTUN2
1	-	10100	UQXTN, UQXTN2
1	0x	10110	FCVTXN, FCVTXN2
1	0x	11010	FCVTNU (vector)
1	0x	11011	FCVTMU (vector)
1	0x	11100	FCVTAU (vector)
1	0x	11101	UCVTF (vector, integer)
1	1x	01100	FCMGE (zero)
1	1x	01101	FCMLE (zero)
1	1x	01110	Unallocated.
1	1x	11010	FCVTPU (vector)
1	1x	11011	FCVTZU (vector, integer)
1	1x	11101	FRSQRTE
1	1x	11111	Unallocated.

Advanced SIMD scalar pairwise

This section describes the encoding of the Advanced SIMD scalar pairwise instruction class. This section is decoded from [Data Processing -- Scalar Floating-Point and Advanced SIMD](#) on page C4-261.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	12	11	10	9	5	4	0
0	1	U	1	1	1	1	0	size	1	1	0	0	0	opcode	1	0	Rn					Rd

Decode fields			Instruction page	Architecture version
U	size	opcode		
-	-	00xxx	Unallocated.	-
-	-	010xx	Unallocated.	-
-	-	01110	Unallocated.	-
-	-	10xxx	Unallocated.	-

Decode fields			Instruction page	Architecture version
U	size	opcode		
-	-	1100x	Unallocated.	-
-	-	11010	Unallocated.	-
-	-	111xx	Unallocated.	-
-	1x	01101	Unallocated.	-
0	-	11011	ADDP (scalar)	-
0	00	01100	FMAXNMP (scalar) - Half-precision variant on page C7-1178	ARMv8.2
0	00	01101	FADDP (scalar) - Half-precision variant on page C7-1055	ARMv8.2
0	00	01111	FMAXP (scalar) - Half-precision variant on page C7-1184	ARMv8.2
0	01	01100	Unallocated.	-
0	01	01101	Unallocated.	-
0	01	01111	Unallocated.	-
0	10	01100	FMINNMP (scalar) - Half-precision variant on page C7-1198	ARMv8.2
0	10	01111	FMINP (scalar) - Half-precision variant on page C7-1204	ARMv8.2
0	11	01100	Unallocated.	-
0	11	01111	Unallocated.	-
1	-	11011	Unallocated.	-
1	0x	01100	FMAXNMP (scalar) - Single-precision and double-precision variant on page C7-1178	-
1	0x	01101	FADDP (scalar) - Single-precision and double-precision variant on page C7-1055	-
1	0x	01111	FMAXP (scalar) - Single-precision and double-precision variant on page C7-1184	-
1	1x	01100	FMINNMP (scalar) - Single-precision and double-precision variant on page C7-1198	-
1	1x	01111	FMINP (scalar) - Single-precision and double-precision variant on page C7-1204	-

Advanced SIMD scalar three different

This section describes the encoding of the Advanced SIMD scalar three different instruction class. This section is decoded from [Data Processing -- Scalar Floating-Point and Advanced SIMD on page C4-261](#).

31	30	29	28	27	26	25	24	23	22	21	20	16			15	12			11	10	9	5		4	0	
0	1	U	1	1	1	1	0	size	1	Rm			opcode			0	0	Rn			Rd					

Decode fields		Instruction page
U	opcode	
-	00xx	Unallocated.
-	01xx	Unallocated.
-	1000	Unallocated.
-	1010	Unallocated.
-	1100	Unallocated.
-	111x	Unallocated.
0	1001	SQDMLAL, SQDMLAL2 (vector)
0	1011	SQDMLSL, SQDMLSL2 (vector)
0	1101	SQDMULL, SQDMULL2 (vector)
1	1001	Unallocated.
1	1011	Unallocated.
1	1101	Unallocated.

Advanced SIMD scalar three same

This section describes the encoding of the Advanced SIMD scalar three same instruction class. This section is decoded from [Data Processing -- Scalar Floating-Point and Advanced SIMD on page C4-261](#).

31	30	29	28	27	26	25	24	23	22	21	20	16			15	11			10	9	5		4	0	
0	1	U	1	1	1	1	0	size	1	Rm			opcode			1	Rn			Rd					

Decode fields			Instruction page
U	size	opcode	
-	-	00000	Unallocated.
-	-	0001x	Unallocated.
-	-	00100	Unallocated.
-	-	011xx	Unallocated.
-	-	1001x	Unallocated.
-	1x	11011	Unallocated.
0	-	00001	SQADD

Decode fields			Instruction page
U	size	opcode	
0	-	00101	SQSUB
0	-	00110	CMGT (register)
0	-	00111	CMGE (register)
0	-	01000	SSHL
0	-	01001	SQSHL (register)
0	-	01010	SRSHL
0	-	01011	SQRSHL
0	-	10000	ADD (vector)
0	-	10001	CMTST
0	-	10100	Unallocated.
0	-	10101	Unallocated.
0	-	10110	SQDMULH (vector)
0	-	10111	Unallocated.
0	0x	11000	Unallocated.
0	0x	11001	Unallocated.
0	0x	11010	Unallocated.
0	0x	11011	FMULX
0	0x	11100	FCMEQ (register)
0	0x	11101	Unallocated.
0	0x	11110	Unallocated.
0	0x	11111	FRECPS
0	1x	11000	Unallocated.
0	1x	11001	Unallocated.
0	1x	11010	Unallocated.
0	1x	11100	Unallocated.
0	1x	11101	Unallocated.
0	1x	11110	Unallocated.
0	1x	11111	FRSQRTS
1	-	00001	UQADD
1	-	00101	UQSUB
1	-	00110	CMHI (register)
1	-	00111	CMHS (register)

Decode fields			Instruction page
U	size	opcode	
1	-	01000	USHL
1	-	01001	UQSHL (register)
1	-	01010	URSHL
1	-	01011	UQRSHL
1	-	10000	SUB (vector)
1	-	10001	CMEQ (register)
1	-	10100	Unallocated.
1	-	10101	Unallocated.
1	-	10110	SQRDMULH (vector)
1	-	10111	Unallocated.
1	0x	11000	Unallocated.
1	0x	11001	Unallocated.
1	0x	11010	Unallocated.
1	0x	11011	Unallocated.
1	0x	11100	FCMGE (register)
1	0x	11101	FACGE
1	0x	11110	Unallocated.
1	0x	11111	Unallocated.
1	1x	11000	Unallocated.
1	1x	11001	Unallocated.
1	1x	11010	FABD
1	1x	11100	FCMGT (register)
1	1x	11101	FACGT
1	1x	11110	Unallocated.
1	1x	11111	Unallocated.

Advanced SIMD scalar shift by immediate

This section describes the encoding of the Advanced SIMD scalar shift by immediate instruction class. This section is decoded from *Data Processing -- Scalar Floating-Point and Advanced SIMD* on page C4-261.

31	30	29	28	27	26	25	24	23	22		19	18	16	15		11	10	9		5	4		0
0	1	U	1	1	1	1	1	0		immh		immb		opcode	1		Rn						Rd

Decode fields

Instruction page

U immh opcode

-	!= 0000	00001	Unallocated.
-	!= 0000	00011	Unallocated.
-	!= 0000	00101	Unallocated.
-	!= 0000	00111	Unallocated.
-	!= 0000	01001	Unallocated.
-	!= 0000	01011	Unallocated.
-	!= 0000	01101	Unallocated.
-	!= 0000	01111	Unallocated.
-	!= 0000	101xx	Unallocated.
-	!= 0000	110xx	Unallocated.
-	!= 0000	11101	Unallocated.
-	!= 0000	11110	Unallocated.
-	0000	-	Unallocated.
0	!= 0000	00000	SSHR
0	!= 0000	00010	SSRA
0	!= 0000	00100	SRSHR
0	!= 0000	00110	SRSRA
0	!= 0000	01000	Unallocated.
0	!= 0000	01010	SHL
0	!= 0000	01100	Unallocated.
0	!= 0000	01110	SQSHL (immediate)
0	!= 0000	10000	Unallocated.
0	!= 0000	10001	Unallocated.
0	!= 0000	10010	SQSHRN, SQSHRN2
0	!= 0000	10011	SQRSHRN, SQRSHRN2
0	!= 0000	11100	SCVTF (vector, fixed-point)
0	!= 0000	11111	FCVTZS (vector, fixed-point)
1	!= 0000	00000	USHR

Decode fields			Instruction page
U	immh	opcode	
1	!= 0000	00010	USRA
1	!= 0000	00100	URSHR
1	!= 0000	00110	URSRA
1	!= 0000	01000	SRI
1	!= 0000	01010	SLI
1	!= 0000	01100	SQSHLU
1	!= 0000	01110	UQSHL (immediate)
1	!= 0000	10000	SQSHRUN, SQSHRUN2
1	!= 0000	10001	SQRSHRUN, SQRSHRUN2
1	!= 0000	10010	UQSHRN, UQSHRN2
1	!= 0000	10011	UQRSHRN, UQRSHRN2
1	!= 0000	11100	UCVTF (vector, fixed-point)
1	!= 0000	11111	FCVTZU (vector, fixed-point)

Advanced SIMD scalar x indexed element

This section describes the encoding of the Advanced SIMD scalar x indexed element instruction class. This section is decoded from *Data Processing -- Scalar Floating-Point and Advanced SIMD* on page C4-261.

31	30	29	28	27	26	25	24	23	22	21	20	19	16	15	12	11	10	9	5	4	0
0	1	U	1	1	1	1	1	size	L	M		Rm		opcode	H	0			Rn		Rd

Decode fields			Instruction page	Architecture version
U	size	opcode		
-	-	0000	Unallocated.	-
-	-	0010	Unallocated.	-
-	-	0100	Unallocated.	-
-	-	0110	Unallocated.	-
-	-	1000	Unallocated.	-
-	-	1010	Unallocated.	-
-	-	1110	Unallocated.	-
-	01	0001	Unallocated.	-
-	01	0101	Unallocated.	-
-	01	1001	Unallocated.	-

Decode fields			Instruction page	Architecture version
U	size	opcode		
0	-	0011	SQDMLAL, SQDMLAL2 (by element)	-
0	-	0111	SQDMLSL, SQDMLSL2 (by element)	-
0	-	1011	SQDMULL, SQDMULL2 (by element)	-
0	-	1100	SQDMULH (by element)	-
0	-	1101	SQRDMULH (by element)	-
0	-	1111	Unallocated.	-
0	00	0001	FMLA (by element) - <i>Scalar, half-precision variant on page C7-1210</i>	ARMv8.2
0	00	0101	FMLS (by element) - <i>Scalar, half-precision variant on page C7-1216</i>	ARMv8.2
0	00	1001	FMUL (by element) - <i>Scalar, half-precision variant on page C7-1233</i>	ARMv8.2
0	1x	0001	FMLA (by element) - <i>Scalar, single-precision and double-precision variant on page C7-1210</i>	-
0	1x	0101	FMLS (by element) - <i>Scalar, single-precision and double-precision variant on page C7-1216</i>	-
0	1x	1001	FMUL (by element) - <i>Scalar, single-precision and double-precision variant on page C7-1233</i>	-
1	-	0011	Unallocated.	-
1	-	0111	Unallocated.	-
1	-	1011	Unallocated.	-
1	-	1100	Unallocated.	-
1	-	1101	SQRDMLAH (by element)	ARMv8.1
1	-	1111	SQRDMLSH (by element)	ARMv8.1
1	00	0001	Unallocated.	-
1	00	0101	Unallocated.	-
1	00	1001	FMULX (by element) - <i>Scalar, half-precision variant on page C7-1241</i>	ARMv8.2
1	1x	0001	Unallocated.	-
1	1x	0101	Unallocated.	-
1	1x	1001	FMULX (by element) - <i>Scalar, single-precision and double-precision variant on page C7-1241</i>	-

Advanced SIMD table lookup

This section describes the encoding of the Advanced SIMD table lookup instruction class. This section is decoded from *Data Processing -- Scalar Floating-Point and Advanced SIMD* on page C4-261.

31	30	29	28	27	26	25	24	23	22	21	20	16	15	14	13	12	11	10	9	5	4	0
0	Q	0	0	1	1	1	0	op2	0	Rm			0	len	op	0	0	Rn			Rd	

Decode fields

Instruction page

op2	len	op	
x1	-	-	Unallocated.
00	00	0	TBL - Single register table variant on page C7-1648
00	00	1	TBX - Single register table variant on page C7-1650
00	01	0	TBL - Two register table variant on page C7-1648
00	01	1	TBX - Two register table variant on page C7-1650
00	10	0	TBL - Three register table variant on page C7-1648
00	10	1	TBX - Three register table variant on page C7-1650
00	11	0	TBL - Four register table variant on page C7-1648
00	11	1	TBX - Four register table variant on page C7-1650
1x	-	-	Unallocated.

Advanced SIMD permute

This section describes the encoding of the Advanced SIMD permute instruction class. This section is decoded from [Data Processing -- Scalar Floating-Point and Advanced SIMD on page C4-261](#).

31	30	29	28	27	26	25	24	23	22	21	20	16	15	14	12	11	10	9	5	4	0
0	Q	0	0	1	1	1	0	size	0	Rm			0	opcode	1	0	Rn			Rd	

Decode fields

Instruction page

opcode

000	Unallocated.
001	UZP1
010	TRN1
011	ZIP1
100	Unallocated.
101	UZP2
110	TRN2
111	ZIP2

Advanced SIMD extract

This section describes the encoding of the Advanced SIMD extract instruction class. This section is decoded from *Data Processing -- Scalar Floating-Point and Advanced SIMD* on page C4-261.

31	30	29	28	27	26	25	24	23	22	21	20	16			15	14	11			10	9	5			4	0	
0	Q	1	0	1	1	1	0	op2	0	Rm			0	imm4			0	Rn			Rd						

Decode fields

Instruction page

op2

x1 Unallocated.

00 [EXT](#)

1x Unallocated.

Advanced SIMD copy

This section describes the encoding of the Advanced SIMD copy instruction class. This section is decoded from *Data Processing -- Scalar Floating-Point and Advanced SIMD* on page C4-261.

31	30	29	28	27	26	25	24	23	22	21	20	16			15	14	11			10	9	5			4	0	
0	Q	op	0	1	1	1	0	0	0	0	imm5			0	imm4			1	Rn			Rd					

Decode fields

Instruction page

Q op imm5 imm4

- - x0000 - Unallocated.

- 0 - 0000 [DUP \(element\)](#)

- 0 - 0001 [DUP \(general\)](#)

- 0 - 0010 Unallocated.

- 0 - 0100 Unallocated.

- 0 - 0110 Unallocated.

- 0 - 1xxx Unallocated.

0 0 - 0011 Unallocated.

0 0 - 0101 [SMOV](#)

0 0 - 0111 [UMOV](#)

0 1 - - Unallocated.

1 0 - 0011 [INS \(general\)](#)

Decode fields				Instruction page
Q	op	imm5	imm4	
1	0	-	0101	SMOV
1	0	x1000	0111	UMOV
1	1	-	-	INS (element)

Advanced SIMD three same (FP16)

This section describes the encoding of the Advanced SIMD three same (FP16) instruction class. This section is decoded from [Data Processing -- Scalar Floating-Point and Advanced SIMD](#) on page C4-261.

31	30	29	28	27	26	25	24	23	22	21	20	16	15	14	13	11	10	9	5	4	0
0	Q	U	0	1	1	1	0	a	1	0		Rm	0	0	opcode	1		Rn			Rd

Decode fields			Instruction page	Architecture version
U	a	opcode		
0	0	000	FMAXNM (vector)	ARMv8.2
0	0	001	FMLA (vector)	ARMv8.2
0	0	010	FADD (vector)	ARMv8.2
0	0	011	FMULX	ARMv8.2
0	0	100	FCMEQ (register)	ARMv8.2
0	0	101	Unallocated.	-
0	0	110	FMAX (vector)	ARMv8.2
0	0	111	FRECPS	ARMv8.2
0	1	000	FMINNM (vector)	ARMv8.2
0	1	001	FMLS (vector)	ARMv8.2
0	1	010	FSUB (vector)	ARMv8.2
0	1	011	Unallocated.	-
0	1	100	Unallocated.	-
0	1	101	Unallocated.	-
0	1	110	FMIN (vector)	ARMv8.2
0	1	111	FRSQRTS	ARMv8.2
1	0	000	FMAXNMP (vector)	ARMv8.2
1	0	001	Unallocated.	-
1	0	010	FADDP (vector)	ARMv8.2
1	0	011	FMUL (vector)	ARMv8.2

Decode fields			Instruction page	Architecture version
U	a	opcode		
1	0	100	FCMGE (register)	ARMv8.2
1	0	101	FACGE	ARMv8.2
1	0	110	FMAXP (vector)	ARMv8.2
1	0	111	FDIV (vector)	ARMv8.2
1	1	000	FMINNMP (vector)	ARMv8.2
1	1	001	Unallocated.	-
1	1	010	FABD	ARMv8.2
1	1	011	Unallocated.	-
1	1	100	FCMGT (register)	ARMv8.2
1	1	101	FACGT	ARMv8.2
1	1	110	FMINP (vector)	ARMv8.2
1	1	111	Unallocated.	-

Advanced SIMD two-register miscellaneous (FP16)

This section describes the encoding of the Advanced SIMD two-register miscellaneous (FP16) instruction class. This section is decoded from *Data Processing -- Scalar Floating-Point and Advanced SIMD* on page C4-261.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	12	11	10	9	5	4	0
0	Q	U	0	1	1	1	0	a	1	1	1	0	0	opcode			1	0	Rn		Rd	

Decode fields			Instruction page	Architecture version
U	a	opcode		
-	-	00xxx	Unallocated.	-
-	-	010xx	Unallocated.	-
-	-	10xxx	Unallocated.	-
-	-	11110	Unallocated.	-
-	0	011xx	Unallocated.	-
-	0	11111	Unallocated.	-
-	1	11100	Unallocated.	-
0	0	11000	FRINTN (vector)	ARMv8.2
0	0	11001	FRINTM (vector)	ARMv8.2
0	0	11010	FCVTNS (vector)	ARMv8.2
0	0	11011	FCVTMS (vector)	ARMv8.2

Decode fields			Instruction page	Architecture version
U	a	opcode		
0	0	11100	FCVTAS (vector)	ARMv8.2
0	0	11101	SCVTF (vector, integer)	ARMv8.2
0	1	01100	FCMGT (zero)	ARMv8.2
0	1	01101	FCMEQ (zero)	ARMv8.2
0	1	01110	FCMLT (zero)	ARMv8.2
0	1	01111	FABS (vector)	ARMv8.2
0	1	11000	FRINTP (vector)	ARMv8.2
0	1	11001	FRINTZ (vector)	ARMv8.2
0	1	11010	FCVTPS (vector)	ARMv8.2
0	1	11011	FCVTZS (vector, integer)	ARMv8.2
0	1	11101	FRECPE	ARMv8.2
0	1	11111	Unallocated.	-
1	0	11000	FRINTA (vector)	ARMv8.2
1	0	11001	FRINTX (vector)	ARMv8.2
1	0	11010	FCVTNU (vector)	ARMv8.2
1	0	11011	FCVTMU (vector)	ARMv8.2
1	0	11100	FCVTAU (vector)	ARMv8.2
1	0	11101	UCVTF (vector, integer)	ARMv8.2
1	1	01100	FCMGE (zero)	ARMv8.2
1	1	01101	FCMLE (zero)	ARMv8.2
1	1	01110	Unallocated.	-
1	1	01111	FNEG (vector)	ARMv8.2
1	1	11000	Unallocated.	-
1	1	11001	FRINTI (vector)	ARMv8.2
1	1	11010	FCVTPU (vector)	ARMv8.2
1	1	11011	FCVTZU (vector, integer)	ARMv8.2
1	1	11101	FRSQRT	ARMv8.2
1	1	11111	FSQRT (vector)	ARMv8.2

Advanced SIMD three same extra

This section describes the encoding of the Advanced SIMD three same extra instruction class. This section is decoded from *Data Processing -- Scalar Floating-Point and Advanced SIMD* on page C4-261.

31	30	29	28	27	26	25	24	23	22	21	20	16			15	14	11			10	9	5			4	0	
0	Q	U	0	1	1	1	0	size	0	Rm			1	opcode			1	Rn			Rd						

Decode fields		Instruction page	Architecture version
U	opcode		
-	001x	Unallocated.	-
-	01xx	Unallocated.	-
0	0000	Unallocated.	-
0	0001	Unallocated.	-
0	1xxx	Unallocated.	-
1	0000	SQRDMLAH (vector)	ARMv8.1
1	0001	SQRDMLSH (vector)	ARMv8.1
1	11x1	Unallocated.	-

Advanced SIMD two-register miscellaneous

This section describes the encoding of the Advanced SIMD two-register miscellaneous instruction class. This section is decoded from [Data Processing -- Scalar Floating-Point and Advanced SIMD](#) on page C4-261.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	12	11	10	9	5	4	0
0	Q	U	0	1	1	1	0	size	1	0	0	0	0	opcode	1	0	Rn		Rd			

Decode fields			Instruction page
U	size	opcode	
-	-	1000x	Unallocated.
-	-	10101	Unallocated.
-	-	11110	Unallocated.
-	0x	011xx	Unallocated.
-	0x	11111	Unallocated.
-	1x	10110	Unallocated.
-	1x	10111	Unallocated.
0	-	00000	REV64
0	-	00001	REV16 (vector)
0	-	00010	SADDLP
0	-	00011	SUQADD

Decode fields			Instruction page
U	size	opcode	
0	-	00100	CLS (vector)
0	-	00101	CNT
0	-	00110	SADALP
0	-	00111	SQABS
0	-	01000	CMGT (zero)
0	-	01001	CMEQ (zero)
0	-	01010	CMLT (zero)
0	-	01011	ABS
0	-	10010	XTN, XTN2
0	-	10011	Unallocated.
0	-	10100	SQXTN, SQXTN2
0	0x	10110	FCVTN, FCVTN2
0	0x	10111	FCVTL, FCVTL2
0	0x	11000	FRINTN (vector)
0	0x	11001	FRINTM (vector)
0	0x	11010	FCVTNS (vector)
0	0x	11011	FCVTMS (vector)
0	0x	11100	FCVTAS (vector)
0	0x	11101	SCVTF (vector, integer)
0	1x	01100	FCMGT (zero)
0	1x	01101	FCMEQ (zero)
0	1x	01110	FCMLT (zero)
0	1x	01111	FABS (vector)
0	1x	11000	FRINTP (vector)
0	1x	11001	FRINTZ (vector)
0	1x	11010	FCVTPS (vector)
0	1x	11011	FCVTZS (vector, integer)
0	1x	11100	URECPE
0	1x	11101	FRECPE
0	1x	11111	Unallocated.
1	-	00000	REV32 (vector)
1	-	00001	Unallocated.

Decode fields			Instruction page
U	size	opcode	
1	-	00010	UADDLP
1	-	00011	USQADD
1	-	00100	CLZ (vector)
1	-	00110	UADALP
1	-	00111	SQNEG
1	-	01000	CMGE (zero)
1	-	01001	CMLE (zero)
1	-	01010	Unallocated.
1	-	01011	NEG (vector)
1	-	10010	SQXTUN, SQXTUN2
1	-	10011	SHLL, SHLL2
1	-	10100	UQXTN, UQXTN2
1	0x	10110	FCVTXN, FCVTXN2
1	0x	10111	Unallocated.
1	0x	11000	FRINTA (vector)
1	0x	11001	FRINTX (vector)
1	0x	11010	FCVTNU (vector)
1	0x	11011	FCVTMU (vector)
1	0x	11100	FCVTAU (vector)
1	0x	11101	UCVTF (vector, integer)
1	00	00101	NOT
1	01	00101	RBIT (vector)
1	1x	00101	Unallocated.
1	1x	01100	FCMGE (zero)
1	1x	01101	FCMLE (zero)
1	1x	01110	Unallocated.
1	1x	01111	FNEG (vector)
1	1x	11000	Unallocated.
1	1x	11001	FRINTI (vector)
1	1x	11010	FCVTPU (vector)
1	1x	11011	FCVTZU (vector, integer)

Decode fields			Instruction page
U	size	opcode	
1	1x	11100	URSQRTE
1	1x	11101	FRSQRTE
1	1x	11111	FSQRT (vector)

Advanced SIMD across lanes

This section describes the encoding of the Advanced SIMD across lanes instruction class. This section is decoded from *Data Processing -- Scalar Floating-Point and Advanced SIMD* on page C4-261.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	12	11	10	9	5	4	0
0	Q	U	0	1	1	1	0	size	1	1	0	0	0	opcode	1	0	Rn					Rd

Decode fields			Instruction page	Architecture version
U	size	opcode		
-	-	0000x	Unallocated.	-
-	-	00010	Unallocated.	-
-	-	001xx	Unallocated.	-
-	-	0100x	Unallocated.	-
-	-	01011	Unallocated.	-
-	-	01101	Unallocated.	-
-	-	01110	Unallocated.	-
-	-	10xxx	Unallocated.	-
-	-	1100x	Unallocated.	-
-	-	111xx	Unallocated.	-
0	-	00011	SADDLV	-
0	-	01010	SMAV	-
0	-	11010	SINV	-
0	-	11011	ADDV	-
0	00	01100	FMAXNMV - <i>Half-precision variant on page C7-1182</i>	ARMv8.2
0	00	01111	FMAV - <i>Half-precision variant on page C7-1188</i>	ARMv8.2
0	01	01100	Unallocated.	-
0	01	01111	Unallocated.	-
0	10	01100	FMINNMV - <i>Half-precision variant on page C7-1202</i>	ARMv8.2
0	10	01111	FMINV - <i>Half-precision variant on page C7-1208</i>	ARMv8.2

Decode fields			Instruction page	Architecture version
U	size	opcode		
0	11	01100	Unallocated.	-
0	11	01111	Unallocated.	-
1	-	00011	UADDLV	-
1	-	01010	UMAXV	-
1	-	11010	UMINV	-
1	-	11011	Unallocated.	-
1	0x	01100	FMAXNMV - <i>Single-precision and double-precision variant on page C7-1182</i>	-
1	0x	01111	FMAXV - <i>Single-precision and double-precision variant on page C7-1188</i>	-
1	1x	01100	FMINNMV - <i>Single-precision and double-precision variant on page C7-1202</i>	-
1	1x	01111	FMINV - <i>Single-precision and double-precision variant on page C7-1208</i>	-

Advanced SIMD three different

This section describes the encoding of the Advanced SIMD three different instruction class. This section is decoded from [Data Processing -- Scalar Floating-Point and Advanced SIMD on page C4-261](#).

31	30	29	28	27	26	25	24	23	22	21	20	16	15	12	11	10	9	5	4	0
0	Q	U	0	1	1	1	0	size	1	Rm	opcode	0	0	Rn	Rd					

Decode fields			Instruction page
U	opcode		
-	1111		Unallocated.
0	0000		SADDL, SADDL2
0	0001		SADDW, SADDW2
0	0010		SSUBL, SSUBL2
0	0011		SSUBW, SSUBW2
0	0100		ADDHN, ADDHN2
0	0101		SABAL, SABAL2
0	0110		SUBHN, SUBHN2
0	0111		SABDL, SABDL2
0	1000		SMLAL, SMLAL2 (vector)
0	1001		SQDMLAL, SQDMLAL2 (vector)
0	1010		SMLSL, SMLSL2 (vector)
0	1011		SQDMLSL, SQDMLSL2 (vector)

Decode fields		Instruction page
U	opcode	
0	1100	SMULL, SMULL2 (vector)
0	1101	SQDMULL, SQDMULL2 (vector)
0	1110	PMULL, PMULL2
1	0000	UADDL, UADDL2
1	0001	UADDW, UADDW2
1	0010	USUBL, USUBL2
1	0011	USUBW, USUBW2
1	0100	RADDHN, RADDHN2
1	0101	UABAL, UABAL2
1	0110	RSUBHN, RSUBHN2
1	0111	UABDL, UABDL2
1	1000	UMLAL, UMLAL2 (vector)
1	1001	Unallocated.
1	1010	UMLSL, UMLSL2 (vector)
1	1011	Unallocated.
1	1100	UMULL, UMULL2 (vector)
1	1101	Unallocated.
1	1110	Unallocated.

Advanced SIMD three same

This section describes the encoding of the Advanced SIMD three same instruction class. This section is decoded from [Data Processing -- Scalar Floating-Point and Advanced SIMD on page C4-261](#).

31 30 29 28				27 26 25 24				23 22 21 20				16 15				11 10 9			5 4		0					
0	Q	U	0	1	1	1	0	size	1	Rm				opcode				1	Rn				Rd			

Decode fields			Instruction page
U	size	opcode	
0	-	00000	SHADD
0	-	00001	SQADD
0	-	00010	SRHADD
0	-	00100	SHSUB
0	-	00101	SQSUB

Decode fields			Instruction page
U	size	opcode	
0	-	00110	CMGT (register)
0	-	00111	CMGE (register)
0	-	01000	SSHL
0	-	01001	SQSHL (register)
0	-	01010	SRSHL
0	-	01011	SQRSHL
0	-	01100	SMAX
0	-	01101	SMIN
0	-	01110	SABD
0	-	01111	SABA
0	-	10000	ADD (vector)
0	-	10001	CMTST
0	-	10010	MLA (vector)
0	-	10011	MUL (vector)
0	-	10100	SMAXP
0	-	10101	SMINP
0	-	10110	SQDMULH (vector)
0	-	10111	ADDP (vector)
0	0x	11000	FMAXNM (vector)
0	0x	11001	FMLA (vector)
0	0x	11010	FADD (vector)
0	0x	11011	FMULX
0	0x	11100	FCMEQ (register)
0	0x	11101	Unallocated.
0	0x	11110	FMAX (vector)
0	0x	11111	FRECPS
0	00	00011	AND (vector)
0	01	00011	BIC (vector, register)
0	1x	11000	FMINNM (vector)
0	1x	11001	FMLS (vector)
0	1x	11010	FSUB (vector)
0	1x	11011	Unallocated.

Decode fields			Instruction page
U	size	opcode	
0	1x	11100	Unallocated.
0	1x	11101	Unallocated.
0	1x	11110	FMIN (vector)
0	1x	11111	FRSQRTS
0	10	00011	ORR (vector, register)
0	11	00011	ORN (vector)
1	-	00000	UHADD
1	-	00001	UQADD
1	-	00010	URHADD
1	-	00100	UHSUB
1	-	00101	UQSUB
1	-	00110	CMHI (register)
1	-	00111	CMHS (register)
1	-	01000	USHL
1	-	01001	UQSHL (register)
1	-	01010	URSHL
1	-	01011	UQRSHL
1	-	01100	UMAX
1	-	01101	UMIN
1	-	01110	UABD
1	-	01111	UABA
1	-	10000	SUB (vector)
1	-	10001	CMEQ (register)
1	-	10010	MLS (vector)
1	-	10011	PMUL
1	-	10100	UMAXP
1	-	10101	UMINP
1	-	10110	SQRDMULH (vector)
1	-	10111	Unallocated.
1	0x	11000	FMAXNMP (vector)
1	0x	11001	Unallocated.
1	0x	11010	FADDP (vector)

Decode fields			Instruction page
U	size	opcode	
1	0x	11011	FMUL (vector)
1	0x	11100	FCMGE (register)
1	0x	11101	FACGE
1	0x	11110	FMAXP (vector)
1	0x	11111	FDIV (vector)
1	00	00011	EOR (vector)
1	01	00011	BSL
1	1x	11000	FMINNMP (vector)
1	1x	11001	Unallocated.
1	1x	11010	FABD
1	1x	11011	Unallocated.
1	1x	11100	FCMGT (register)
1	1x	11101	FACGT
1	1x	11110	FMINP (vector)
1	1x	11111	Unallocated.
1	10	00011	BIT
1	11	00011	BIF

Advanced SIMD modified immediate

This section describes the encoding of the Advanced SIMD modified immediate instruction class. This section is decoded from [Data Processing -- Scalar Floating-Point and Advanced SIMD on page C4-261](#).

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	12	11	10	9	8	7	6	5	4	0
0	Q	op	0	1	1	1	1	0	0	0	0	a	b	c	cmode		o2	1	d	e	f	g	h	Rd		

Decode fields				Instruction page	Architecture version
Q	op	cmode	o2		
-	0	0xxx	1	Unallocated.	-
-	0	0xx0	0	MOVI - 32-bit shifted immediate variant on page C7-1385	-
-	0	0xx1	0	ORR (vector, immediate) - 32-bit variant on page C7-1399	-
-	0	10xx	1	Unallocated.	-
-	0	10x0	0	MOVI - 16-bit shifted immediate variant on page C7-1385	-
-	0	10x1	0	ORR (vector, immediate) - 16-bit variant on page C7-1399	-

Decode fields				Instruction page	Architecture version
Q	op	cmode	o2		
-	0	110x	0	MOVI - 32-bit shifting ones variant on page C7-1385	-
-	0	110x	1	Unallocated.	-
-	0	1110	0	MOVI - 8-bit variant on page C7-1385	-
-	0	1110	1	Unallocated.	-
-	0	1111	0	FMOV (vector, immediate) - Single-precision variant on page C7-1222	-
-	0	1111	1	FMOV (vector, immediate) - Half-precision variant on page C7-1222	ARMv8.2
-	1	-	1	Unallocated.	-
-	1	0xx0	0	MVNI - 32-bit shifted immediate variant on page C7-1393	-
-	1	0xx1	0	BIC (vector, immediate) - 32-bit variant on page C7-995	-
-	1	10x0	0	MVNI - 16-bit shifted immediate variant on page C7-1393	-
-	1	10x1	0	BIC (vector, immediate) - 16-bit variant on page C7-995	-
-	1	110x	0	MVNI - 32-bit shifting ones variant on page C7-1393	-
0	1	1110	0	MOVI - 64-bit scalar variant on page C7-1385	-
0	1	1111	0	Unallocated.	-
1	1	1110	0	MOVI - 64-bit vector variant on page C7-1385	-
1	1	1111	0	FMOV (vector, immediate) - Double-precision variant on page C7-1222	-

Advanced SIMD shift by immediate

This section describes the encoding of the Advanced SIMD shift by immediate instruction class. This section is decoded from [Data Processing -- Scalar Floating-Point and Advanced SIMD on page C4-261](#).

31	30	29	28	27	26	25	24	23	22	19	18	16	15	11	10	9	5	4	0
0	Q	U	0	1	1	1	1	0	!=0000	immb	opcode	1	Rn	Rd					
immh																			

Decode fields		Instruction page
U	opcode	
-	00001	Unallocated.
-	00011	Unallocated.
-	00101	Unallocated.
-	00111	Unallocated.
-	01001	Unallocated.
-	01011	Unallocated.
-	01101	Unallocated.

Decode fields		Instruction page
U	opcode	
-	01111	Unallocated.
-	10101	Unallocated.
-	1011x	Unallocated.
-	110xx	Unallocated.
-	11101	Unallocated.
-	11110	Unallocated.
0	00000	SSHR
0	00010	SSRA
0	00100	SRSHR
0	00110	SRSRA
0	01000	Unallocated.
0	01010	SHL
0	01100	Unallocated.
0	01110	SQSHL (immediate)
0	10000	SHRN, SHRN2
0	10001	RSHRN, RSHRN2
0	10010	SQSHRN, SQSHRN2
0	10011	SQRSHRN, SQRSHRN2
0	10100	SSHLL, SSHLL2
0	11100	SCVTF (vector, fixed-point)
0	11111	FCVTZS (vector, fixed-point)
1	00000	USHR
1	00010	USRA
1	00100	URSHR
1	00110	URSRA
1	01000	SRI
1	01010	SLI
1	01100	SQSHLU
1	01110	UQSHL (immediate)
1	10000	SQSHRUN, SQSHRUN2
1	10001	SQRSHRUN, SQRSHRUN2
1	10010	UQSHRN, UQSHRN2

Decode fields		
U	opcode	Instruction page
1	10011	UQRSHRN, UQRSHRN2
1	10100	USHLL, USHLL2
1	11100	UCVTF (vector, fixed-point)
1	11111	FCVTZU (vector, fixed-point)

Advanced SIMD vector x indexed element

This section describes the encoding of the Advanced SIMD vector x indexed element instruction class. This section is decoded from *Data Processing -- Scalar Floating-Point and Advanced SIMD on page C4-261*.

31	30	29	28	27	26	25	24	23	22	21	20	19	16	15	12	11	10	9	5	4	0
0	Q	U	0	1	1	1	1	size	L	M		Rm	opcode	H	0		Rn			Rd	

Decode fields			Instruction page	Architecture version
U	size	opcode		
-	-	1110	Unallocated.	-
-	01	1001	Unallocated.	-
0	-	0000	Unallocated.	-
0	-	0010	SMLAL, SMLAL2 (by element)	-
0	-	0011	SQDMLAL, SQDMLAL2 (by element)	-
0	-	0100	Unallocated.	-
0	-	0110	SMLSL, SMLSL2 (by element)	-
0	-	0111	SQDMLSL, SQDMLSL2 (by element)	-
0	-	1000	MUL (by element)	-
0	-	1010	SMULL, SMULL2 (by element)	-
0	-	1011	SQDMULL, SQDMULL2 (by element)	-
0	-	1100	SQDMULH (by element)	-
0	-	1101	SQRDMULH (by element)	-
0	-	1111	Unallocated.	-
0	00	0001	FMLA (by element) - <i>Vector, half-precision variant on page C7-1211</i>	ARMv8.2
0	00	0101	FMLS (by element) - <i>Vector, half-precision variant on page C7-1217</i>	ARMv8.2
0	00	1001	FMUL (by element) - <i>Vector, half-precision variant on page C7-1234</i>	ARMv8.2
0	01	0001	Unallocated.	-
0	01	0101	Unallocated.	-

Decode fields			Instruction page	Architecture version
U	size	opcode		
0	1x	0001	FMLA (by element) - <i>Vector, single-precision and double-precision variant on page C7-1211</i>	-
0	1x	0101	FMLS (by element) - <i>Vector, single-precision and double-precision variant on page C7-1217</i>	-
0	1x	1001	FMUL (by element) - <i>Vector, single-precision and double-precision variant on page C7-1234</i>	-
1	-	0000	MLA (by element)	-
1	-	0010	UMLAL, UMLAL2 (by element)	-
1	-	0100	MLS (by element)	-
1	-	0110	UMLSL, UMLSL2 (by element)	-
1	-	1000	Unallocated.	-
1	-	1010	UMULL, UMULL2 (by element)	-
1	-	1011	Unallocated.	-
1	-	1100	Unallocated.	-
1	-	1101	SQRDMLAH (by element)	ARMv8.1
1	-	1111	SQRDMLSH (by element)	ARMv8.1
1	00	0001	Unallocated.	-
1	00	0011	Unallocated.	-
1	00	0101	Unallocated.	-
1	00	0111	Unallocated.	-
1	00	1001	FMULX (by element) - <i>Vector, half-precision variant on page C7-1242</i>	ARMv8.2
1	1x	0101	Unallocated.	-
1	1x	1001	FMULX (by element) - <i>Vector, single-precision and double-precision variant on page C7-1242</i>	-
1	11	0001	Unallocated.	-
1	11	0011	Unallocated.	-
1	11	0111	Unallocated.	-

Conversion between floating-point and fixed-point

This section describes the encoding of the Conversion between floating-point and fixed-point instruction class. This section is decoded from *Data Processing -- Scalar Floating-Point and Advanced SIMD* on page C4-261.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	16	15			10	9			5	4			0
sf	0	S	1	1	1	1	0	type	0	rmode	opcode	scale				Rn				Rd						

Decode fields							Instruction page	Architecture version
sf	S	type	rmode	opcode	scale			
-	-	-	-	1xx	-		Unallocated.	-
-	-	-	x0	00x	-		Unallocated.	-
-	-	-	x1	01x	-		Unallocated.	-
-	-	-	0x	00x	-		Unallocated.	-
-	-	-	1x	01x	-		Unallocated.	-
-	-	10	-	-	-		Unallocated.	-
-	1	-	-	-	-		Unallocated.	-
0	-	-	-	-	0xxxxx		Unallocated.	-
0	0	00	00	010	-		SCVTF (scalar, fixed-point) - 32-bit to single-precision variant on page C7-1443	-
0	0	00	00	011	-		UCVTF (scalar, fixed-point) - 32-bit to single-precision variant on page C7-1680	-
0	0	00	11	000	-		FCVTZS (scalar, fixed-point) - Single-precision to 32-bit variant on page C7-1150	-
0	0	00	11	001	-		FCVTZU (scalar, fixed-point) - Single-precision to 32-bit variant on page C7-1160	-
0	0	01	00	010	-		SCVTF (scalar, fixed-point) - 32-bit to double-precision variant on page C7-1443	-
0	0	01	00	011	-		UCVTF (scalar, fixed-point) - 32-bit to double-precision variant on page C7-1680	-
0	0	01	11	000	-		FCVTZS (scalar, fixed-point) - Double-precision to 32-bit variant on page C7-1150	-
0	0	01	11	001	-		FCVTZU (scalar, fixed-point) - Double-precision to 32-bit variant on page C7-1160	-
0	0	11	00	010	-		SCVTF (scalar, fixed-point) - 32-bit to half-precision variant on page C7-1443	ARMv8.2
0	0	11	00	011	-		UCVTF (scalar, fixed-point) - 32-bit to half-precision variant on page C7-1680	ARMv8.2
0	0	11	11	000	-		FCVTZS (scalar, fixed-point) - Half-precision to 32-bit variant on page C7-1150	ARMv8.2
0	0	11	11	001	-		FCVTZU (scalar, fixed-point) - Half-precision to 32-bit variant on page C7-1160	ARMv8.2
1	0	00	00	010	-		SCVTF (scalar, fixed-point) - 64-bit to single-precision variant on page C7-1443	-

Decode fields						Instruction page	Architecture version
sf	S	type	rmode	opcode	scale		
1	0	00	00	011	-	UCVTF (scalar, fixed-point) - <i>64-bit to single-precision variant on page C7-1680</i>	-
1	0	00	11	000	-	FCVTZS (scalar, fixed-point) - <i>Single-precision to 64-bit variant on page C7-1150</i>	-
1	0	00	11	001	-	FCVTZU (scalar, fixed-point) - <i>Single-precision to 64-bit variant on page C7-1160</i>	-
1	0	01	00	010	-	SCVTF (scalar, fixed-point) - <i>64-bit to double-precision variant on page C7-1443</i>	-
1	0	01	00	011	-	UCVTF (scalar, fixed-point) - <i>64-bit to double-precision variant on page C7-1680</i>	-
1	0	01	11	000	-	FCVTZS (scalar, fixed-point) - <i>Double-precision to 64-bit variant on page C7-1150</i>	-
1	0	01	11	001	-	FCVTZU (scalar, fixed-point) - <i>Double-precision to 64-bit variant on page C7-1160</i>	-
1	0	11	00	010	-	SCVTF (scalar, fixed-point) - <i>64-bit to half-precision variant on page C7-1443</i>	ARMv8.2
1	0	11	00	011	-	UCVTF (scalar, fixed-point) - <i>64-bit to half-precision variant on page C7-1680</i>	ARMv8.2
1	0	11	11	000	-	FCVTZS (scalar, fixed-point) - <i>Half-precision to 64-bit variant on page C7-1150</i>	ARMv8.2
1	0	11	11	001	-	FCVTZU (scalar, fixed-point) - <i>Half-precision to 64-bit variant on page C7-1160</i>	ARMv8.2

Conversion between floating-point and integer

This section describes the encoding of the Conversion between floating-point and integer instruction class. This section is decoded from *Data Processing -- Scalar Floating-Point and Advanced SIMD on page C4-261*.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	16	15	14	13	12	11	10	9	5	4	0
sf	0	S	1	1	1	1	0	type	1	rmode	opcode	0	0	0	0	0	0	0	Rn					Rd

Decode fields						Instruction page	Architecture version
sf	S	type	rmode	opcode			
-	-	-	x1	01x		Unallocated.	-
-	-	-	x1	10x		Unallocated.	-
-	-	-	1x	01x		Unallocated.	-
-	-	-	1x	10x		Unallocated.	-
-	0	10	-	0xx		Unallocated.	-

Decode fields					Instruction page	Architecture version
sf	S	type	rmode	opcode		
-	0	10	-	10x	Unallocated.	-
-	1	-	-	-	Unallocated.	-
0	0	00	x1	11x	Unallocated.	-
0	0	00	00	000	FCVTNS (scalar) - Single-precision to 32-bit variant on page C7-1125	-
0	0	00	00	001	FCVTNU (scalar) - Single-precision to 32-bit variant on page C7-1130	-
0	0	00	00	010	SCVTF (scalar, integer) - 32-bit to single-precision variant on page C7-1445	-
0	0	00	00	011	UCVTF (scalar, integer) - 32-bit to single-precision variant on page C7-1682	-
0	0	00	00	100	FCVTAS (scalar) - Single-precision to 32-bit variant on page C7-1101	-
0	0	00	00	101	FCVTAU (scalar) - Single-precision to 32-bit variant on page C7-1106	-
0	0	00	00	110	FMOV (general) - Single-precision to 32-bit variant on page C7-1226	-
0	0	00	00	111	FMOV (general) - 32-bit to single-precision variant on page C7-1226	-
0	0	00	01	000	FCVTPS (scalar) - Single-precision to 32-bit variant on page C7-1135	-
0	0	00	01	001	FCVTPU (scalar) - Single-precision to 32-bit variant on page C7-1140	-
0	0	00	1x	11x	Unallocated.	-
0	0	00	10	000	FCVTMS (scalar) - Single-precision to 32-bit variant on page C7-1113	-
0	0	00	10	001	FCVTMU (scalar) - Single-precision to 32-bit variant on page C7-1118	-
0	0	00	11	000	FCVTZS (scalar, integer) - Single-precision to 32-bit variant on page C7-1152	-
0	0	00	11	001	FCVTZU (scalar, integer) - Single-precision to 32-bit variant on page C7-1162	-
0	0	01	0x	11x	Unallocated.	-
0	0	01	00	000	FCVTNS (scalar) - Double-precision to 32-bit variant on page C7-1125	-
0	0	01	00	001	FCVTNU (scalar) - Double-precision to 32-bit variant on page C7-1130	-
0	0	01	00	010	SCVTF (scalar, integer) - 32-bit to double-precision variant on page C7-1445	-

Decode fields					Instruction page	Architecture version
sf	S	type	rmode	opcode		
0	0	01	00	011	UCVTF (scalar, integer) - <i>32-bit to double-precision variant on page C7-1682</i>	-
0	0	01	00	100	FCVTAS (scalar) - <i>Double-precision to 32-bit variant on page C7-1101</i>	-
0	0	01	00	101	FCVTAU (scalar) - <i>Double-precision to 32-bit variant on page C7-1106</i>	-
0	0	01	01	000	FCVTPS (scalar) - <i>Double-precision to 32-bit variant on page C7-1135</i>	-
0	0	01	01	001	FCVTPU (scalar) - <i>Double-precision to 32-bit variant on page C7-1140</i>	-
0	0	01	10	000	FCVTMS (scalar) - <i>Double-precision to 32-bit variant on page C7-1113</i>	-
0	0	01	10	001	FCVTMU (scalar) - <i>Double-precision to 32-bit variant on page C7-1118</i>	-
0	0	01	10	11x	Unallocated.	-
0	0	01	11	000	FCVTZS (scalar, integer) - <i>Double-precision to 32-bit variant on page C7-1152</i>	-
0	0	01	11	001	FCVTZU (scalar, integer) - <i>Double-precision to 32-bit variant on page C7-1162</i>	-
0	0	01	11	111	Unallocated.	-
0	0	10	-	11x	Unallocated.	-
0	0	11	00	000	FCVTNS (scalar) - <i>Half-precision to 32-bit variant on page C7-1125</i>	ARMv8.2
0	0	11	00	001	FCVTNU (scalar) - <i>Half-precision to 32-bit variant on page C7-1130</i>	ARMv8.2
0	0	11	00	010	SCVTF (scalar, integer) - <i>32-bit to half-precision variant on page C7-1445</i>	ARMv8.2
0	0	11	00	011	UCVTF (scalar, integer) - <i>32-bit to half-precision variant on page C7-1682</i>	ARMv8.2
0	0	11	00	100	FCVTAS (scalar) - <i>Half-precision to 32-bit variant on page C7-1101</i>	ARMv8.2
0	0	11	00	101	FCVTAU (scalar) - <i>Half-precision to 32-bit variant on page C7-1106</i>	ARMv8.2
0	0	11	00	110	FMOV (general) - <i>Half-precision to 32-bit variant on page C7-1226</i>	ARMv8.2
0	0	11	00	111	FMOV (general) - <i>32-bit to half-precision variant on page C7-1226</i>	ARMv8.2
0	0	11	01	000	FCVTPS (scalar) - <i>Half-precision to 32-bit variant on page C7-1135</i>	ARMv8.2

Decode fields					Instruction page	Architecture version
sf	S	type	rmode	opcode		
0	0	11	01	001	FCVTPU (scalar) - <i>Half-precision to 32-bit variant on page C7-1140</i>	ARMv8.2
0	0	11	10	000	FCVTMS (scalar) - <i>Half-precision to 32-bit variant on page C7-1113</i>	ARMv8.2
0	0	11	10	001	FCVTMU (scalar) - <i>Half-precision to 32-bit variant on page C7-1118</i>	ARMv8.2
0	0	11	11	000	FCVTZS (scalar, integer) - <i>Half-precision to 32-bit variant on page C7-1152</i>	ARMv8.2
0	0	11	11	001	FCVTZU (scalar, integer) - <i>Half-precision to 32-bit variant on page C7-1162</i>	ARMv8.2
1	0	00	-	11x	Unallocated.	-
1	0	00	00	000	FCVTNS (scalar) - <i>Single-precision to 64-bit variant on page C7-1125</i>	-
1	0	00	00	001	FCVTNU (scalar) - <i>Single-precision to 64-bit variant on page C7-1130</i>	-
1	0	00	00	010	SCVTF (scalar, integer) - <i>64-bit to single-precision variant on page C7-1445</i>	-
1	0	00	00	011	UCVTF (scalar, integer) - <i>64-bit to single-precision variant on page C7-1682</i>	-
1	0	00	00	100	FCVTAS (scalar) - <i>Single-precision to 64-bit variant on page C7-1101</i>	-
1	0	00	00	101	FCVTAU (scalar) - <i>Single-precision to 64-bit variant on page C7-1106</i>	-
1	0	00	01	000	FCVTPS (scalar) - <i>Single-precision to 64-bit variant on page C7-1135</i>	-
1	0	00	01	001	FCVTPU (scalar) - <i>Single-precision to 64-bit variant on page C7-1140</i>	-
1	0	00	10	000	FCVTMS (scalar) - <i>Single-precision to 64-bit variant on page C7-1113</i>	-
1	0	00	10	001	FCVTMU (scalar) - <i>Single-precision to 64-bit variant on page C7-1118</i>	-
1	0	00	11	000	FCVTZS (scalar, integer) - <i>Single-precision to 64-bit variant on page C7-1152</i>	-
1	0	00	11	001	FCVTZU (scalar, integer) - <i>Single-precision to 64-bit variant on page C7-1162</i>	-
1	0	01	x1	11x	Unallocated.	-
1	0	01	00	000	FCVTNS (scalar) - <i>Double-precision to 64-bit variant on page C7-1125</i>	-
1	0	01	00	001	FCVTNU (scalar) - <i>Double-precision to 64-bit variant on page C7-1130</i>	-

Decode fields					Instruction page	Architecture version
sf	S	type	rmode	opcode		
1	0	01	00	010	SCVTF (scalar, integer) - <i>64-bit to double-precision variant on page C7-1445</i>	-
1	0	01	00	011	UCVTF (scalar, integer) - <i>64-bit to double-precision variant on page C7-1682</i>	-
1	0	01	00	100	FCVTAS (scalar) - <i>Double-precision to 64-bit variant on page C7-1101</i>	-
1	0	01	00	101	FCVTAU (scalar) - <i>Double-precision to 64-bit variant on page C7-1106</i>	-
1	0	01	00	110	FMOV (general) - <i>Double-precision to 64-bit variant on page C7-1227</i>	-
1	0	01	00	111	FMOV (general) - <i>64-bit to double-precision variant on page C7-1226</i>	-
1	0	01	01	000	FCVTPS (scalar) - <i>Double-precision to 64-bit variant on page C7-1135</i>	-
1	0	01	01	001	FCVTPU (scalar) - <i>Double-precision to 64-bit variant on page C7-1140</i>	-
1	0	01	1x	11x	Unallocated.	-
1	0	01	10	000	FCVTMS (scalar) - <i>Double-precision to 64-bit variant on page C7-1113</i>	-
1	0	01	10	001	FCVTMU (scalar) - <i>Double-precision to 64-bit variant on page C7-1118</i>	-
1	0	01	11	000	FCVTZS (scalar, integer) - <i>Double-precision to 64-bit variant on page C7-1152</i>	-
1	0	01	11	001	FCVTZU (scalar, integer) - <i>Double-precision to 64-bit variant on page C7-1162</i>	-
1	0	10	x0	11x	Unallocated.	-
1	0	10	01	110	FMOV (general) - <i>Top half of 128-bit to 64-bit variant on page C7-1227</i>	-
1	0	10	01	111	FMOV (general) - <i>64-bit to top half of 128-bit variant on page C7-1226</i>	-
1	0	10	1x	11x	Unallocated.	-
1	0	11	00	000	FCVTNS (scalar) - <i>Half-precision to 64-bit variant on page C7-1125</i>	ARMv8.2
1	0	11	00	001	FCVTNU (scalar) - <i>Half-precision to 64-bit variant on page C7-1130</i>	ARMv8.2
1	0	11	00	010	SCVTF (scalar, integer) - <i>64-bit to half-precision variant on page C7-1445</i>	ARMv8.2
1	0	11	00	011	UCVTF (scalar, integer) - <i>64-bit to half-precision variant on page C7-1682</i>	ARMv8.2

Decode fields					Instruction page	Architecture version
sf	S	type	rmode	opcode		
1	0	11	00	100	FCVTAS (scalar) - Half-precision to 64-bit variant on page C7-1101	ARMv8.2
1	0	11	00	101	FCVTAU (scalar) - Half-precision to 64-bit variant on page C7-1106	ARMv8.2
1	0	11	00	110	FMOV (general) - Half-precision to 64-bit variant on page C7-1226	ARMv8.2
1	0	11	00	111	FMOV (general) - 64-bit to half-precision variant on page C7-1226	ARMv8.2
1	0	11	01	000	FCVTPS (scalar) - Half-precision to 64-bit variant on page C7-1135	ARMv8.2
1	0	11	01	001	FCVTPU (scalar) - Half-precision to 64-bit variant on page C7-1140	ARMv8.2
1	0	11	10	000	FCVTMS (scalar) - Half-precision to 64-bit variant on page C7-1113	ARMv8.2
1	0	11	10	001	FCVTMU (scalar) - Half-precision to 64-bit variant on page C7-1118	ARMv8.2
1	0	11	11	000	FCVTZS (scalar, integer) - Half-precision to 64-bit variant on page C7-1152	ARMv8.2
1	0	11	11	001	FCVTZU (scalar, integer) - Half-precision to 64-bit variant on page C7-1162	ARMv8.2

Floating-point data-processing (1 source)

This section describes the encoding of the Floating-point data-processing (1 source) instruction class. This section is decoded from [Data Processing -- Scalar Floating-Point and Advanced SIMD on page C4-261](#).

31	30	29	28	27	26	25	24	23	22	21	20	15	14	13	12	11	10	9	5	4	0
M	0	S	1	1	1	1	0	type	1	opcode				1	0	0	0	0	Rn		Rd

Decode fields				Instruction page	Architecture version
M	S	type	opcode		
-	-	-	x1xxxx	Unallocated.	-
-	-	-	1xxxxx	Unallocated.	-
-	1	-	-	Unallocated.	-
0	0	00	000000	FMOV (register) - Single-precision variant on page C7-1224	-
0	0	00	000001	FABS (scalar) - Single-precision variant on page C7-1041	-
0	0	00	000010	FNEG (scalar) - Single-precision variant on page C7-1250	-
0	0	00	000011	FSQRT (scalar) - Single-precision variant on page C7-1302	-

Decode fields				Instruction page	Architecture version
M	S	type	opcode		
0	0	00	000100	Unallocated.	-
0	0	00	000101	FCVT - Single-precision to double-precision variant on page C7-1096	-
0	0	00	000110	Unallocated.	-
0	0	00	000111	FCVT - Single-precision to half-precision variant on page C7-1096	-
0	0	00	001000	FRINTN (scalar) - Single-precision variant on page C7-1280	-
0	0	00	001001	FRINTP (scalar) - Single-precision variant on page C7-1284	-
0	0	00	001010	FRINTM (scalar) - Single-precision variant on page C7-1276	-
0	0	00	001011	FRINTZ (scalar) - Single-precision variant on page C7-1292	-
0	0	00	001100	FRINTA (scalar) - Single-precision variant on page C7-1268	-
0	0	00	001101	Unallocated.	-
0	0	00	001110	FRINTX (scalar) - Single-precision variant on page C7-1288	-
0	0	00	001111	FRINTI (scalar) - Single-precision variant on page C7-1272	-
0	0	01	000000	FMOV (register) - Double-precision variant on page C7-1224	-
0	0	01	000001	FABS (scalar) - Double-precision variant on page C7-1041	-
0	0	01	000010	FNEG (scalar) - Double-precision variant on page C7-1250	-
0	0	01	000011	FSQRT (scalar) - Double-precision variant on page C7-1302	-
0	0	01	000100	FCVT - Double-precision to single-precision variant on page C7-1096	-
0	0	01	000101	Unallocated.	-
0	0	01	000110	Unallocated.	-
0	0	01	000111	FCVT - Double-precision to half-precision variant on page C7-1096	-
0	0	01	001000	FRINTN (scalar) - Double-precision variant on page C7-1280	-
0	0	01	001001	FRINTP (scalar) - Double-precision variant on page C7-1284	-
0	0	01	001010	FRINTM (scalar) - Double-precision variant on page C7-1276	-
0	0	01	001011	FRINTZ (scalar) - Double-precision variant on page C7-1292	-
0	0	01	001100	FRINTA (scalar) - Double-precision variant on page C7-1268	-
0	0	01	001101	Unallocated.	-
0	0	01	001110	FRINTX (scalar) - Double-precision variant on page C7-1288	-
0	0	01	001111	FRINTI (scalar) - Double-precision variant on page C7-1272	-
0	0	10	00xxxx	Unallocated.	-
0	0	11	000000	FMOV (register) - Half-precision variant on page C7-1224	ARMv8.2
0	0	11	000001	FABS (scalar) - Half-precision variant on page C7-1041	ARMv8.2
0	0	11	000010	FNEG (scalar) - Half-precision variant on page C7-1250	ARMv8.2

Decode fields				Instruction page	Architecture version
M	S	type	opcode		
0	0	11	000011	FSQRT (scalar) - Half-precision variant on page C7-1302	ARMv8.2
0	0	11	000100	FCVT - Half-precision to single-precision variant on page C7-1096	-
0	0	11	000101	FCVT - Half-precision to double-precision variant on page C7-1096	-
0	0	11	00011x	Unallocated.	-
0	0	11	001000	FRINTN (scalar) - Half-precision variant on page C7-1280	ARMv8.2
0	0	11	001001	FRINTP (scalar) - Half-precision variant on page C7-1284	ARMv8.2
0	0	11	001010	FRINTM (scalar) - Half-precision variant on page C7-1276	ARMv8.2
0	0	11	001011	FRINTZ (scalar) - Half-precision variant on page C7-1292	ARMv8.2
0	0	11	001100	FRINTA (scalar) - Half-precision variant on page C7-1268	ARMv8.2
0	0	11	001101	Unallocated.	-
0	0	11	001110	FRINTX (scalar) - Half-precision variant on page C7-1288	ARMv8.2
0	0	11	001111	FRINTI (scalar) - Half-precision variant on page C7-1272	ARMv8.2
1	-	-	-	Unallocated.	-

Floating-point compare

This section describes the encoding of the Floating-point compare instruction class. This section is decoded from [Data Processing -- Scalar Floating-Point and Advanced SIMD on page C4-261](#).

31 30 29 28				27 26 25 24				23 22 21 20				16 15 14 13 12 11 10 9								5 4		0				
M	0	S	1	1	1	1	0	type	1	Rm				op	1	0	0	0	Rn				opcode2			

Decode fields					Instruction page	Architecture version
M	S	type	op	opcode2		
-	-	-	-	xxxx1	Unallocated.	-
-	-	-	-	xxx1x	Unallocated.	-
-	-	-	-	xx1xx	Unallocated.	-
-	-	-	x1	-	Unallocated.	-
-	-	-	1x	-	Unallocated.	-
-	-	10	-	-	Unallocated.	-
-	1	-	-	-	Unallocated.	-
0	0	00	00	00000	FCMP	-
0	0	00	00	01000	FCMP	-
0	0	00	00	10000	FCMPE	-

Decode fields					Instruction page	Architecture version
M	S	type	op	opcode2		
0	0	00	00	11000	FCMPE	-
0	0	01	00	00000	FCMP	-
0	0	01	00	01000	FCMP	-
0	0	01	00	10000	FCMPE	-
0	0	01	00	11000	FCMPE	-
0	0	11	00	00000	FCMP	ARMv8.2
0	0	11	00	01000	FCMP	ARMv8.2
0	0	11	00	10000	FCMPE	ARMv8.2
0	0	11	00	11000	FCMPE	ARMv8.2
1	-	-	-	-	Unallocated.	-

Floating-point immediate

This section describes the encoding of the Floating-point immediate instruction class. This section is decoded from *Data Processing -- Scalar Floating-Point and Advanced SIMD* on page C4-261.

31 30 29 28 27 26 25 24 23 22 21 20														13 12 11 10 9						5 4				0	
M	0	S	1	1	1	1	0	type	1	imm8				1	0	0	imm5			Rd					

Decode fields				Instruction page	Architecture version
M	S	type	imm5		
-	-	-	xxxx1	Unallocated.	-
-	-	-	xxx1x	Unallocated.	-
-	-	-	xx1xx	Unallocated.	-
-	-	-	x1xxx	Unallocated.	-
-	-	-	1xxxx	Unallocated.	-
-	-	10	-	Unallocated.	-
-	1	-	-	Unallocated.	-
0	0	00	00000	FMOV (scalar, immediate) - <i>Single-precision variant</i> on page C7-1229	-
0	0	01	00000	FMOV (scalar, immediate) - <i>Double-precision variant</i> on page C7-1229	-
0	0	11	00000	FMOV (scalar, immediate) - <i>Half-precision variant</i> on page C7-1229	ARMv8.2
1	-	-	-	Unallocated.	-

Floating-point conditional compare

This section describes the encoding of the Floating-point conditional compare instruction class. This section is decoded from *Data Processing -- Scalar Floating-Point and Advanced SIMD* on page C4-261.

31	30	29	28	27	26	25	24	23	22	21	20	16	15	12	11	10	9	5	4	3	0
M	0	S	1	1	1	1	0	type	1	Rm	cond	0	1	Rn	op	nzc	v				

Decode fields				Instruction page	Architecture version
M	S	type	op		
-	-	10	-	Unallocated.	-
-	1	-	-	Unallocated.	-
0	0	00	0	FCCMP - Single-precision variant on page C7-1059	-
0	0	00	1	FCCMPE - Single-precision variant on page C7-1061	-
0	0	01	0	FCCMP - Double-precision variant on page C7-1059	-
0	0	01	1	FCCMPE - Double-precision variant on page C7-1061	-
0	0	11	0	FCCMP - Half-precision variant on page C7-1059	ARMv8.2
0	0	11	1	FCCMPE - Half-precision variant on page C7-1061	ARMv8.2
1	-	-	-	Unallocated.	-

Floating-point data-processing (2 source)

This section describes the encoding of the Floating-point data-processing (2 source) instruction class. This section is decoded from *Data Processing -- Scalar Floating-Point and Advanced SIMD* on page C4-261.

31	30	29	28	27	26	25	24	23	22	21	20	16	15	12	11	10	9	5	4	0
M	0	S	1	1	1	1	0	type	1	Rm	opcode	1	0	Rn	Rd					

Decode fields				Instruction page	Architecture version
M	S	type	opcode		
-	-	-	1xx1	Unallocated.	-
-	-	-	1x1x	Unallocated.	-
-	-	-	11xx	Unallocated.	-
-	-	10	-	Unallocated.	-
-	1	-	-	Unallocated.	-
0	0	00	0000	FMUL (scalar) - Single-precision variant on page C7-1239	-
0	0	00	0001	FDIV (scalar) - Single-precision variant on page C7-1166	-
0	0	00	0010	FADD (scalar) - Single-precision variant on page C7-1053	-

Decode fields				Instruction page	Architecture version
M	S	type	opcode		
0	0	00	0011	FSUB (scalar) - <i>Single-precision variant on page C7-1306</i>	-
0	0	00	0100	FMAX (scalar) - <i>Single-precision variant on page C7-1172</i>	-
0	0	00	0101	FMIN (scalar) - <i>Single-precision variant on page C7-1192</i>	-
0	0	00	0110	FMAXNM (scalar) - <i>Single-precision variant on page C7-1176</i>	-
0	0	00	0111	FMINNM (scalar) - <i>Single-precision variant on page C7-1196</i>	-
0	0	00	1000	FNMUL (scalar) - <i>Single-precision variant on page C7-1256</i>	-
0	0	01	0000	FMUL (scalar) - <i>Double-precision variant on page C7-1239</i>	-
0	0	01	0001	FDIV (scalar) - <i>Double-precision variant on page C7-1166</i>	-
0	0	01	0010	FADD (scalar) - <i>Double-precision variant on page C7-1053</i>	-
0	0	01	0011	FSUB (scalar) - <i>Double-precision variant on page C7-1306</i>	-
0	0	01	0100	FMAX (scalar) - <i>Double-precision variant on page C7-1172</i>	-
0	0	01	0101	FMIN (scalar) - <i>Double-precision variant on page C7-1192</i>	-
0	0	01	0110	FMAXNM (scalar) - <i>Double-precision variant on page C7-1176</i>	-
0	0	01	0111	FMINNM (scalar) - <i>Double-precision variant on page C7-1196</i>	-
0	0	01	1000	FNMUL (scalar) - <i>Double-precision variant on page C7-1256</i>	-
0	0	11	0000	FMUL (scalar) - <i>Half-precision variant on page C7-1239</i>	ARMv8.2
0	0	11	0001	FDIV (scalar) - <i>Half-precision variant on page C7-1166</i>	ARMv8.2
0	0	11	0010	FADD (scalar) - <i>Half-precision variant on page C7-1053</i>	ARMv8.2
0	0	11	0011	FSUB (scalar) - <i>Half-precision variant on page C7-1306</i>	ARMv8.2
0	0	11	0100	FMAX (scalar) - <i>Half-precision variant on page C7-1172</i>	ARMv8.2
0	0	11	0101	FMIN (scalar) - <i>Half-precision variant on page C7-1192</i>	ARMv8.2
0	0	11	0110	FMAXNM (scalar) - <i>Half-precision variant on page C7-1176</i>	ARMv8.2
0	0	11	0111	FMINNM (scalar) - <i>Half-precision variant on page C7-1196</i>	ARMv8.2
0	0	11	1000	FNMUL (scalar) - <i>Half-precision variant on page C7-1256</i>	ARMv8.2
1	-	-	-	Unallocated.	-

Floating-point conditional select

This section describes the encoding of the Floating-point conditional select instruction class. This section is decoded from *Data Processing -- Scalar Floating-Point and Advanced SIMD* on page C4-261.

31	30	29	28	27	26	25	24	23	22	21	20	16	15	12	11	10	9	5	4	0
M	0	S	1	1	1	1	0	type	1		Rm		cond	1	1		Rn			Rd

Decode fields			Instruction page	Architecture version
M	S	type		
-	-	10	Unallocated.	-
-	1	-	Unallocated.	-
0	0	00	FCSEL - Single-precision variant on page C7-1094	-
0	0	01	FCSEL - Double-precision variant on page C7-1094	-
0	0	11	FCSEL - Half-precision variant on page C7-1094	ARMv8.2
1	-	-	Unallocated.	-

Floating-point data-processing (3 source)

This section describes the encoding of the Floating-point data-processing (3 source) instruction class. This section is decoded from [Data Processing -- Scalar Floating-Point and Advanced SIMD on page C4-261](#).

31	30	29	28	27	26	25	24	23	22	21	20	16	15	14	10	9	5	4	0
M	0	S	1	1	1	1	1	type	o1		Rm	o0		Ra		Rn			Rd

Decode fields					Instruction page	Architecture version
M	S	type	o1	o0		
-	-	10	-	-	Unallocated.	-
-	1	-	-	-	Unallocated.	-
0	0	00	0	0	FMADD - Single-precision variant on page C7-1168	-
0	0	00	0	1	FMSUB - Single-precision variant on page C7-1231	-
0	0	00	1	0	FNMADD - Single-precision variant on page C7-1252	-
0	0	00	1	1	FNMSUB - Single-precision variant on page C7-1254	-
0	0	01	0	0	FMADD - Double-precision variant on page C7-1168	-
0	0	01	0	1	FMSUB - Double-precision variant on page C7-1231	-
0	0	01	1	0	FNMADD - Double-precision variant on page C7-1252	-
0	0	01	1	1	FNMSUB - Double-precision variant on page C7-1254	-
0	0	11	0	0	FMADD - Half-precision variant on page C7-1168	ARMv8.2
0	0	11	0	1	FMSUB - Half-precision variant on page C7-1231	ARMv8.2

Decode fields					Instruction page	Architecture version
M	S	type	o1	o0		
0	0	11	1	0	FNMADD - Half-precision variant on page C7-1252	ARMv8.2
0	0	11	1	1	FNMSUB - Half-precision variant on page C7-1254	ARMv8.2
1	-	-	-	-	Unallocated.	-

